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Internship report

Stochastic Treatment of the Formation of Ice Mantles on Interstellar Dust Grains

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Alone, what did Bloom feel?

The cold of interstellar space, thousands of degrees below freezing point or the absolute zero of Fahrenheit, Centigrade or Réaumur: the incipient intimations of proximate dawn.

James Joyce, *Ulysses*

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Foreword

The present report results from an internship I made from March to June 2023 at the *Laboratoire d'Etude du Rayonnement et de la Matière en Astrophysique* (LERMA) in Meudon, under the supervision of Franck Le Petit and Emeric Bron, to graduate with a Master's Degree in Astronomy and Astrophysics at the Observatoire de Paris.

The topic of the internship was the study of surface processes on interstellar grains when accounting for the strong temperature fluctuations that maintain small grains out of equilibrium. The precise goal was to better understand the impact of such fluctuations on the formation of ice mantles on grains. Using the framework developed by Emeric Bron during his thesis [1], I built a simple model for the formation of ice mantles on grains stochastically heated by stellar UV photons based off the master equation approach. I developed a standalone Fortran code from scratch called `StoCapy`[1] to compute the probability distributions of grain temperature and populations. I highlighted regimes for which the stochastic treatment invalidated the results obtained by the classical and deterministic rate equation approach, and built approximations to palliate the errors without resorting to the numerically costly computations of the statistical framework.

Cosmic rays were then included in the model, to understand the desorption of ice mantle species triggered by the transient heating of the grain that they too can induce. This was based on the values of energies absorbed by grains upon impact with a cosmic ray computed by Kalvans et al in a series of paper [2, 3, 4]. We took into account the cooling of the grain caused by the evaporation of large amounts of its mantle, which was no longer negligible compared to the usual radiative cooling. This additional coupling between temperature and surface population made the mathematical problem significantly more complex and required the implementation of a new numerical method. We used our results to assess different rate equation prescriptions of the literature. We showed that they fail to accurately reproduce the desorption caused by cosmic rays on the full range of gas conditions. However, we derived a refined approximation to replace the full stochastic computation, which should be fast enough to be implemented in astrochemical models.

This report is divided to follow the chronology of the internship: the first section introduces the astrophysical question as well as the two considered methods: the tools from stochastic processes that we use and the rate equations to which we compare our results. The second section showcases the work done in the first half of the internship: the derivation of a stochastic model for the formation of ice mantles, its resolution and the results. The third section covers the addition of cosmic rays to the model, it highlights some of the difficulties encountered in the resolution, presents the results and their approximations. The last section is a brief conclusion on the work done during the internship. We discuss the limits of our results and end on some prospects for further studies.

The appendices are dedicated to the lengthy computations that we will skip in the presentation, to the details of the numerical methods, and to additional figures.

1. Introduction

1.1. Astrophysical context

Far from empty, the interstellar space is a complex medium displaying an impressive variety of physical processes and a complex chemistry. At the end of an intricate cycle of cooling and collapse, the formation of new stars takes place in the denser parts of the Interstellar Medium (ISM). The interstellar space is not only filled with gas but with a variety of small solid particles. Although it only represents a percent of the mass of gas, interstellar dust is at the heart of many of the mechanisms that grant the ISM its astrophysical richness. Dust absorbs ultraviolet (UV) light and hence contributes to shielding molecules from photo-dissociating photons. These molecules participate in the cooling of the gas, via the emission of photons in their ro-vibrational levels. The cold gas can then eventually collapse and form new stars. Dust is also responsible for the heating of the gas through the photo-

[1] For lack of higher inspirations, the code was named to combine the word *stochastic* and the recent obsession of the author with capybaras.

electric effect, as UV photons eject energetic electrons from the grains. Finally, and most importantly for the subject of this report, the surface of the dust grains is also a catalyst for the formation of many chemical species. A complete overview of the properties of dust and its many roles in the ISM is out of the scope of this short introduction (we refer a curious reader to the reference book by Draine [5]). In the sequel, our focus will be on the central role held by dust in the chemistry of the ISM.

Key aspects of interstellar chemistry take place on the surfaces of dust grains. Particles in the gas can collide with a grain and be adsorbed on its surface, in adsorption sites. Hopping or tunnelling from site to site on the surface, they can eventually meet each other and react. Then, the newly formed molecules can desorb from the grain to enrich the gas in diverse species. This offers a pathway for many reactions that are otherwise slow and inefficient in the gas phase. Not only does this make surface chemistry an undeniable factor in the chemical richness of the ISM, but its implications reach beyond, to broader and still not fully tackled questions.

First, among the species formed on grains, molecular hydrogen (H_2) is the most abundant molecule of the ISM [6, 7, 8]. Understanding in which form is the hydrogen in the ISM is crucial because it dictates the boundary between atomic and molecular gas, and only the latter can efficiently cool down and collapse to form stars. Secondly, in the deeper molecular parts of the ISM, species can accumulate on the cold surface. It leads to the formation of ice mantles around the grain core, of which main constituents are H_2O, CO, CO2, NH_3 [9]. It is important to understand the formation of these mantles, since they are eventually brought to prestellar environment and protoplanetary disks, and could later influence the composition of planetoids and planets [10]. These mantles can also help grains merge together into macroscopic formations [11]. Third, Complex Organic Molecules are assumed to be formed in ices, knowing whether these molecules are related to the origin of life or not remains a hot topic [12]. In view of all this, an accurate modelling of the surface processes on grains is key to the understanding of not only the ISM chemistry, but of the formation of stars, of planets, and of the life cycle of the ISM as a whole.

The present report focuses on two questions in the study of ice mantles around dust grains. On the one hand, we will consider strongly UV-illuminated environments, typically photo-dissociation regions (PDR) at the edge of clouds [13, 14] or protoplanetary disks [15]. In these regions, the exposition of the grains to the intense radiation maintains them at high temperatures and prevents the formation of ice. The depth at which the grains become cold enough and ice mantles are not evaporated away is the so-called *ice line*: its role is important in the complex chemistry of PDRs, which is highly dependent on depth. The PDRs in turn provide information on the feedback of newly formed stars on the ISM. In young stellar objects, the position of ice lines determines the structure and evolution of protoplanetary disks [16, 17]: ice participates in grain agglomeration to eventually form planets, and the species trapped in the ice can survive the intense radiation that progressively cleanses the disk. As we will discuss throughout this report, the absorption of photons makes the temperature of small dust grains fluctuate. The first question we wish to answer in this work is how taking into account these temperature fluctuations affects the photo-evaporation of ice mantles in UV-illuminated regions.

On the other hand, deep in molecular clouds, the dust temperature gets extremely low. Without other desorbing processes than evaporation, the gas would entirely freeze-out on the grains. However, many species are actually observed in the gas phase with, among them, molecules that are only formed in ices, such as methanol. This is a testimony that other desorption mechanisms, often dubbed as *non thermal* processes, are at play. Several of these non-thermal desorption processes have been studied, and none has proven to be everywhere dominant, nor to explain perfectly the observed abundances [18]. First, UV photons can strip molecules off the grain in so-called photodesorption [19], but deep inside clouds, only the secondary UV photons[2] remain and their flux is low. Second, the energy of exothermic reactions occurring at the surface could also be used to expel material of the mantle, as observed in the experiments done at LERMA [20, 21]. Finally, the cosmic rays permeating the ISM can hit a grain and produce a transient high-temperature spike [22]. At the point of impact, a hot spot is created from which molecules desorb, in a process referred to as *sputtering* [18, 23]. The

[2]Produced by the interaction of cosmic rays with the molecular hydrogen gas.

heat then diffuses to the entire grain and raises its temperature, boosting thermal desorption for a short time. This latter process is known as Whole Grain Heating [24, 25]. This is the second question in our work: we seek to know how and how much the transient heating of the grain by cosmic rays triggers the evaporation of molecules from the ice mantles.

With an aim at answering both questions, we develop a grain model for ice mantles that we first apply to the formation of ice in strongly illuminated environments. In a second time, we refine the model to include the desorption induced by cosmic-rays. As chemical reactions are out of the scope of this work in view of the time available, pure ice mantles of a single species are considered. We choose to study H_2O and CO ices, as they are among the most abundant ice constituents [9]. This choice was made to consider a refractory species (H_2O) and a volatile one (CO). Both of these species are central in molecular astrochemistry, and models must accurately reproduce their abundances to interpret the observations (as both H_2O and CO can be directly observed in gas and ice phases, and are used to infer information on other less observable species). Our grain model will be relying on a stochastic formalism that accounts for the randomness of photons and cosmic rays hitting the grain. The rest of this introduction is dedicated to motivating the resort to such a method, and presenting the mathematical stochastic framework.

1.2. Modelling grain surfaces

1.2.1. Rate equations

Surface chemistry is as important in the ISM as it is complex to model: behind the simple picture depicted above of species adsorbing, reacting, and returning to the gas or forming ice mantles, hide numerous micro-physical processes [26]. Different levels of approximation have been implemented in numerical models to emulate these processes.

To simulate interstellar chemistry, most models resort to the so-called formalism of *rate equations*: the tracked quantities are the densities $n(\mathrm{A})$ (in particle per unit volume) for each chemical species A. These quantities are averaged over the whole gas, or the whole dust population (adsorbed and gas-phase species being usually treated as two distinct quantities), in a mean-field approximation. Hasegawa, Herbst and Leung [27] were among the first to develop gas-phase and grain-surface simulations of large chemistry networks, and some of their prescriptions for modelling many processes are still widely spread today (whereas other ones have been updated, see [28] for a review). Typically, in these models, hundreds of species are considered, with thousands of reactions in the network. The evolution of $n(\mathrm{A})$ over time is governed by an ordinary differential equation (ODE), where $\frac{dn(\mathrm{A})}{dt}$ is given by the total average rate of creation of A per second minus the total average destruction rate of A. As an illustration, consider a molecule A adsorbed on a grain, that can be created by adsorption of its gas counterpart, denoted A_{gas}, and that can be destructed either by reacting with B, or by desorbing from the grain. Its rate equation would then write:

$$\frac{dn(\mathrm{A})}{dt} = k_{ads} n(\mathrm{A}_{gas}) - k_{des} n(\mathrm{A}) - k_{\mathrm{A,B}} n(\mathrm{A}) n(\mathrm{B}). \tag{1}$$

The above terms are intuitively understood: if each adsorbed molecule A_{gas} has a equal probability k_{ads} of adsorbing per second, the total mean adsorption should be proportional to $n(\mathrm{A}_{gas})$, and similarly for the desorption of A from the surface. The rate of two-body reactions between A and B is dictated by the rate of collisions between the two and is thus expected to be proportional to the product $n(\mathrm{A})n(\mathrm{B})$.

Most of the physical information is contained in the adsorption, desorption and reaction rate coefficients k_i, and the solutions heavily depend on their values. However, these coefficients are not constants, and in fact display an exponential dependence on the dust temperature T_d of the form $k_i \propto \exp(-\frac{E}{kT_d})$ where E is a characteristic energy of the process. Grain temperature influences the desorption rate of surface species, their mobility on the surface, and the reactions rates that present a chemical barrier. Most astrochemical models assume thermal equilibrium of the grain (defined by the balance between the radiative power absorbed by the grain and the radiative power it emits),

thus fixing its temperature and the value of the rate coefficients[3]. In doing so, these models completely ignore possible temperature fluctuations that can affect exponentially-dependent rates. Yet, if thermal equilibrium might hold for large grains, it is not the case of smaller ones.

1.2.2. *Stochastic fluctuations*

It is well-documented that smaller grains (of lesser radius than $\lesssim 20\mathrm{nm}$) do not have a constant temperature. Indeed, they are so light that their typical thermal energy is lower than or comparable to the energy of a UV photon (see [29, 30], and section 24.2 in [5]). This implies that no stationary equilibrium is possible, as the reception of a single photon significantly increases the grain's energy content, provoking a temperature spike. The grain then cools down to temperature lowers than equilibrium before absorbing another UV photon. Monte Carlo simulations of small grains hence have shown that their temperature as a function of time is a succession of spikes at several hundred Kelvin separated by plateaus at low temperatures [28] (see Figure 1). Modelling the surface processes on small grains assuming a constant equilibrium temperature will inevitably miss out on the real outcome of those mechanisms. Furthermore the small grains are the most abundant in the ISM and offer the most available surface, so a significant share of surface chemistry is in fact subject to temperature fluctuations.

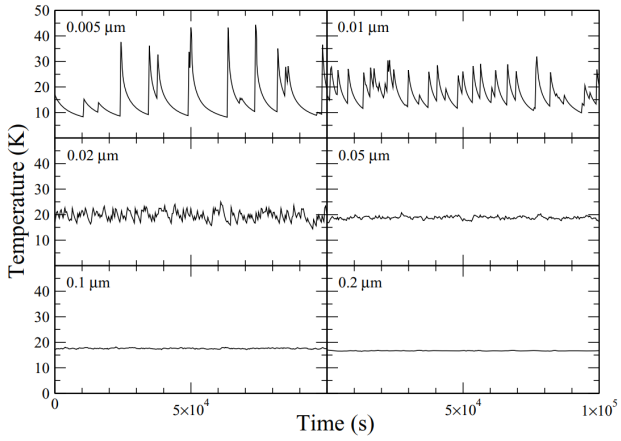


Figure 1: Temperature of interstellar grains of different sizes over time (figure adapted from [28]).

The stochastic nature of temperature has lead some authors to shift from a deterministic approach to propose a statistical treatment of the quantities of interest. Notably, Bron et al. [31] showed that the efficiency of the formation of H_2 on grains is strongly improved by the lower-than-equilibrium temperature that small grains have most of the time, in greater adequacy with observational evidence.

Their stochastic approach is based on the master equation, which treats the temperature and population on the grain as stochastic processes rather than deterministic quantities: the absorption and emission of photons, as well as surface processes (adsorptions, reactions, desorptions) are considered as random Poisson events rather than following a fixed rate.

The master equation approach had been first applied to surface chemistry by Biham et al. [32], to study the effects of the discreteness of the surface population alone (hence with a fixed temperature). Indeed, if the temperature fluctuations will remain the central focus of this work, a number of other physical considerations are in favor of a stochastic, rather than deterministic, approach of surface processes. For instance, if there are on average less than one molecule on the surface of each grain, the important factor controlling chemical reactions is the probability for two reactants to be present at the same time on the surface so that they can meet (this is the *accretion limited* regime), a point completely missed by rate equations such as Eq. (1). In this setting as well, full Monte Carlo astrochemical simulations of have shed light on the inadequacies of rate equations in this regime (see for instance [33, 34]), but they remain computationally costly. The aforementioned master equation approach, as well as moment equations [6] and compromises founded on modified rate equations [35] have been proposed to account for the stochastic effects. However, these promising methods have not yet been applied to their full possible extent.

The goal of this work is to expand the master equation approach to the study of ice mantles, accounting for both the temperature and surface population fluctuations. It will be applied to the two astrophysical questions presented above: first, the stochastic impact of photons at the edge of

[3] Additionally, models often consider a single grain size, rather than a distribution of sizes, yet the temperature of grains is not independent of their size.

clouds and disks, and in a second time the impact of cosmic rays deep in molecular clouds. We close this introduction with the presentation of the stochastic formalism and the master equation.

1.2.3. Mathematical framework: the Master Equation

For many micro-physical processes, the reasonable approach is often to abandon an exact description and turn to the derivation of averaged quantities : this is the mean-field approximation at the heart of rate equations. However, a single mean value is not always meaningful. This is typically the case of the temperature of small grains we discussed above: they receive high energy photons that bring them to a high temperature, then cool down to a low one, before encountering another photon, undergoing another spike, and so on. In practice, they do not spend much time at their equilibrium temperature, and the strongly non-linear dependence of the desorption rate on the temperature means that the average rate can differ strongly from the rate taken at the average temperature. The better modelling choice is to consider the absorption and emission of photons happens *at random*, and to model them as Poisson processes. With this choice, the temperature is not a traditional function of time but becomes a *stochastic process*. The same can be said of the randomness of the adsorption and desorption of species (among other mechanisms), meaning the population can also be considered to be a stochastic process.

For our purpose, a stochastic process is in broad words a random function of time. In our case, the temperature would be a *pure jump stochastic process*, that is to say a piecewise constant function of time which at random times jumps by a random amount. The jumps here correspond to photon absorption and emission. Considering a stochastic process, one is not only interested by its average value, but can try to infer the probability of every possible value, which provides a more complete description. Let us now present the formalism itself, that allows for the derivation of these probabilities.

The Master Equation. Consider a system of different possible states, perpetually kept out of equilibrium by transitions from one state to another, like a small interstellar grain. We need to define how the system jumps from one state to another. The system is modelled by an abstract stochastic process $\mathbb{S}$ that can take values among states denoted X. For grains, X should be described by the relevant quantities of the grain: its temperature and the population on its surface. We wish to know the probability $f(X,t)$ of the event $\mathbb{S}(t) = X$, that is the probability of a grain to be in state X at time t. Let us say that the probability of being in a state $X \in \mathcal{X}$ at $t + dt$ knowing that the system is in state Y at t is written $p_{Y\to X}dt$. The quantity $p_{Y\to X}$ is called the transition rate, in probability per unit time. It contains all the physical processes of the model. If one knows $f(X,t)$ for all X —that is to say the probabilities of all the possible states of the stochastic process $\mathbb{S}$ at time t, based on the transition rate one is able to deduce the probabilities of finding the system at time $t + dt$ in any of those states. This is expressed in the master equation[4], a time differential equation on f which writes:

$$\frac{d}{dt}f(X,t) = \int f(Y,t)p_{Y\to X}dY - f(X,t)\int p_{X\to Y}dY. \tag{2}$$

The interpretation is the following: the evolution of the probability to find the system in state X is the difference between the probability of $\mathbb{S}$ arriving to state X from other states Y per second (the first term), and the probability of $\mathbb{S}$ leaving state X towards other states Y per second (the second term).

Statistical equilibrium. Observing a single grain at several times, it will never have the same temperature. However, if one looks at the many grains in a large volume, the fraction of them that is in a given phase of the cycle should be the same every time, as they are mostly desynchronised and independent from each other. This hypothesis that the proportion of grains in a given state is

[4]The master equation is a special case of the differential Chapman-Kolmogorov equation, derived independently by both in their study of Markov processes. Coincidental piece of trivia, the name *master equation* was actually first proposed in a 1940 paper by Nordsieck, Lamb and Uhlenback, on cosmic-ray showers.

constant over time is called *statistical equilibrium*, and it mathematically says that the PDF f is time-independent. We should thus solve for a stationary solution of the master equation:

$$\int f(Y) p_{Y\to X} dY = f(X) \int p_{X\to Y} dY. \tag{3}$$

This simply expresses the balance between arrivals at state X and departures from state X.

Our formalism relies on the stationary master equation. In the sequel, a state X will be the data of the grain temperature and the surface population. After carefully defining the transition rates $p_{X\to Y}$, one can solve for the distribution of temperatures and surface population of interstellar grains. The difficulties will reside in solving Eq. (3), which is a linear equation but in an infinite-dimensional vector space. Our techniques will be at first mainly adapted from Bron et al. [31] where such equations were numerically solved in the case of H_2 formation on grains. However, as we extend the formalism to the study of ice mantles, we will also derive systems of equations yet unconsidered in the literature that will require new methods.

We are now all set to present the grain model and apply the master equation to the grain's temperature and population.

2. Stochastic impact of photons on the formation of ice mantles

This section is dedicated to the first astrophysical question presented in the introduction: the formation of ice mantles on interstellar dust in strongly illuminated environments of the ISM, such as the edges of clouds and disks. The abundant UV photons lead to fluctuations of the temperature of smaller grains, that we wish to fully take into account by building a grain model based on the master equation. The ice mantles considered are constituted of either CO or H_2O, which are built by adsorbing species from the gas phase. In the intense radiation setting that we consider, the two main possible desorption mechanisms are thermal evaporation, which will be greatly affected by temperature fluctuations, and photodesorption, because of the abundance of high energy photons.

The physical model is first presented, alongside a rate-equation treatment of the problem, which will serve as a comparison ground. We then derive a stochastic model based on the master equation. The resolution of the master equation was implemented numerically and we give a short overview of the numerical methods (most details are relegated to the appendix). We discuss the results and give physical interpretations of the behaviours that rate equations fail to predict. These interpretations allow us to build approximations of the full stochastic computation.

2.1. Physical model

We present our grain model in this section. The two key variables to describe the ice mantles are the number of molecules on the surface, of course, and the grain temperature, both because it is the quantity directly affected by the stochastic nature of photon emission and absorption, and because in turn it regulates the thermal desorption and thus the population on the grain. Thus, most of our attention will focus on modelling the evolution of the grain's thermal energy and surface population. The basis for our model is the one from Bron et al. [31], used to simulate the formation of H_2 on grains, with additions and adaptations to treat ice mantles. The grain's other parameters, such as its geometry, are less of our concern and we will stick to elementary considerations for the sake of simplicity. The grains will be considered spherical, of radius a. The distribution of radii in this work is $1\text{nm} \leq a \leq 300\text{nm}$, but we are mainly interested in smaller grains ($1\text{nm} \leq a \leq 20\text{nm}$) since they are the most subject to wide temperature variations. The grains are assumed to be composed of amorphous carbon, as it is the dominant composition at the sizes of interest (bigger grains being silicates and smaller grains, PAHs, typically).

2.1.1. *Grain temperature and thermal energy*

The grain's temperature reflects its internal thermal energy content, which designates the energy the material can store in the vibration modes of its lattice. The heat capacity C of a material describes the change dE in its energy when its temperature is raised by dT, i.e. $C = \frac{dE}{dT}$. At higher temperatures, new internal modes are possible and the grain typically requires more energy to raise his temperature raise by the same amount. Hence, heat capacity is temperature-dependent, and the relation between dust temperature T_d and grain energy E is the one-to-one relation

$$E(T_d) = \int_0^{T_d} C(T)dT. \tag{4}$$

We derive C from the volumic heat capacity (in $J \cdot K^{-1} \cdot \mathrm{cm}^{-3}$) of amorphous carbon, which behaves mostly as a T^2 power-law at the relevant temperatures (see Figure 2). The heat capacity, and thus the energy, of the full grain at a given temperature then scales with its volume: $E(T_d) \propto a^3$.

We assume that the grain temperature is controlled by radiative processes only: it is heated by the absorption of photons from the interstellar radiation field, and its cooling mechanism is the emission of IR photons. We neglect collisional heating, that only dominates over radiation deeper inside dark clouds. To determine the grain temperature, one only has to track the evolution of its thermal energy using (4). It is much easier to work with energies in radiative processes instead of temperatures: indeed, the absorption of a photon of energy U simply raises the grain thermal energy by the same amount, and similarly the emission by the grain of a photon of energy U amounts to the decrease of E by an energy U.

The ambient radiation field is supposed isotropic. It is described by the intensity $I_U(U)$ (in $W/\mathrm{m}^2/J/sr$, the index U indicates that the intensity is given per interval of photon energy). Because of its small size, the efficiency of absorption of photons by the grain is not simply given by its geometric cross-section, as it would be the case in the macroscopic setting $a \gg \lambda$ (where $\lambda = \frac{hc}{U}$ is the wavelength of a photon of energy U). Instead, the efficiency of photon absorption is characterized by the absorption coefficient, which is the ratio of the photon-absorption cross-section to the geometrical cross-section of the grain. The number of photons having energies between U and $U+dU$ absorbed by the grain per second is given by the mean flux of photons in this energy range, $4\pi\frac{I_U(U)}{U}dU$, multiplied by the photon-absorption cross-section to yield:

$$R_{abs}(U)dU = 4\pi^2 a^2 Q_{abs}(U)\frac{I_U(U)}{U}dU.$$

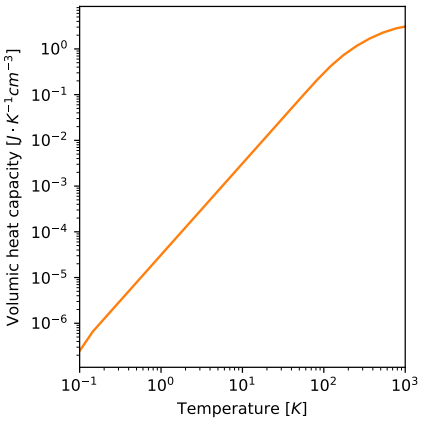


Figure 2: Volumic heat capacity of amorphous carbon.

From this absorption rate (in photons per unit time), one can obtain the mean absorbed power coming from photons in $[U,U+dU]$, by multiplying the rate by the photon energy U : $P_{abs}(U) = UR_{abs}(U)$. We derive a similar expression for the grain cooling. The grain at temperature T_d emits as a modified black body of specific intensity $Q_{abs}(U)B_U(U,T_d)$ ($B_U(U,T_d)$ is the black body intensity at temperature T given by Planck's law, expressed in $W/\mathrm{m}^2/J/sr$). The rate of emission of photons in the energy range $[U,U+dU]$ writes:

$$R_{em}(U,T_d)dU = 4\pi^2 a^2 Q_{abs}(U)\frac{B_U(U,T_d)}{U}dU,$$

and the corresponding emitted power is $P_{em}(U,T_d) = UR_{em}(U,T_d)$. If fluctuations are neglected, the temperature at equilibrium T_d^{eq} is given by the balance of total emitted and absorbed powers:

$$\int_0^{\infty} P_{abs}(U)dU = \int_0^{E(T_d^{eq})} P_{em}(U,T_d^{eq})dU. \tag{5}$$

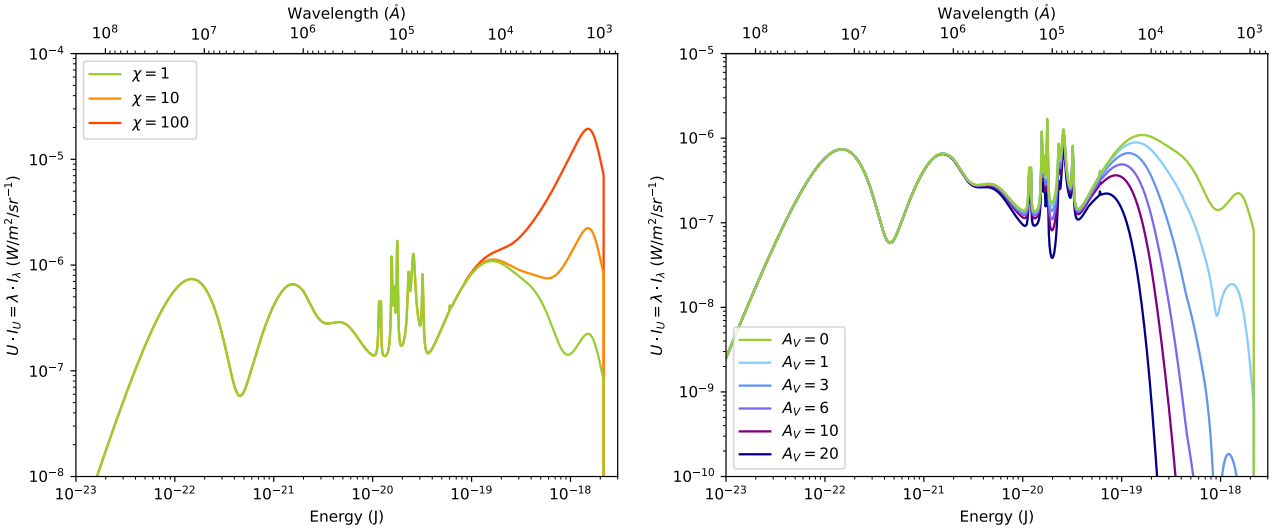


Figure 3: *Left:* Spectrum of the Standard Interstellar Radiation Field ($\chi = 1$) and spectra for $\chi =$ 10,100. *Right:* Spectra of the radiation field for different extinctions A_V ($\chi = 1$).

Here we stress that the right integral is only up to $E(T_d)$, since the grain cannot emit a photon of energy larger than its own. The thermal energy of lighter grains is small enough to become comparable to the photon energies that they could possibly emit, effectively cutting down on their cooling ability.

2.1.2. *Radiation field*

The specific intensity I_U that is responsible for grain heating is that of the standard interstellar radiation field (ISRF). We rely on the ISRF model by Mathis et al. [36]. In this paragraph we briefly summarize its content. It is the sum of four contributions over four spectral domains: The Cosmic Microwave Background (CMB) dominates the larger wavelengths (10^8 to $8 \cdot 10^6 \dot{A}$), it is a black body at temperature $T_{CMB} = 2.73K$. The infrared part ($8 \cdot 10^6$ to $6 \cdot 10^4 \dot{A}$), which is the spectral band presenting the most complex features, is mostly due to dust emission from the surrounding environment. The visible to near-UV part of the spectrum ($6 \cdot 10^4$ to $2450 \dot{A}$) is dominated by starlight. Finally, the far UV emission (2450 to $912 \dot{A}$) is due to massive O and B type stars. The ISRF displays a strong cut at the wavelength $912 \dot{A}$. This is the Lyman limit: at shorter wavelengths, photons ionize atomic hydrogen, so there are no photon at higher energies than the Lyman limit in neutral regions of the ISM.

The far UV part of the radiation field, from $912 \dot{A}$ to $2450 \dot{A}$, is of great importance because its high energy photons are the ones responsible for the biggest temperature spikes of interstellar grains. In strong PDRs such as the Orion bar for instance, this band is much more intense than in other parts of the ISM, because of the proximity of massive stars that illuminate the cloud edge. To model the presence of such stars in cloud edges and disks, and hence the intensity of the far UV component, we scale by a factor χ the far UV contribution of the ISRF. The effect of different χ is shown in Figure 3.

Between the very edge of the cloud and deeper regions, the radiation field gets progressively extinct, mainly by interstellar dust (in addition to the specific absorption lines corresponding to the self-shielding of H_2, but that we do not consider here). This extinction does not affect all wavelengths uniformly and mainly affects the UV and visible spectrum. It is usually quantified by the parameter A_V measuring in magnitudes the ratio of non-extinguished intensity I_λ^0 to extinguished one I_λ, at the frequency $\lambda = 547\text{nm}$:

$$A_V = 2.5 \log \left(\frac{I_\lambda^0}{I_\lambda} \right).$$

We use the model of extinction curves from the Meudon PDR code, which relies on the series of paper by Fitzpatrick [37]. The right panel of Figure 3 displays the radiation field at different extinction values. Remark the (negative) bump at $2175 \dot{A}$ that will play a role later (see [37] for details on this

feature).

We will study the impact of both extinction A_V and scaling factor χ on the grain temperature. As explained above, the extinction A_V models the depth in the cloud, while the scaling factor χ models the intense UV radiation of the environments we consider.

2.1.3. *Surface population*

Now that we covered the temperature of the grain, we describe our model for the ice mantle around the grain. The composition of the mantle is fixed to a single species A (we will consider A to be H_2O and CO in our applications). This means we only allow molecules of A from the gas phase to adsorb onto the grain. If enough molecules are adsorbed, they fill up the surface and start to form several layers of ice. One can easily foresee how geometrically intricate a full treatment of the surface could become, as the ice can form complex porous structures. However, our goal is only to estimate the size of the mantle through the number N of A molecules on the grain, so we will make simple assumptions.

The process is modelled as follows: we suppose that a layer of ice on the grain is composed of N_s adsorption sites to which molecules can bind. The number of sites N_s on one layer is proportional to the grain surface: we fix $N_s = 4\pi a^2 d_s^{-2}$ where d_s is a characteristic distance between sites (its numerical value is given in Table 1). We suppose that the molecules adsorb and desorb in an orderly, layer-by-layer fashion, so these processes always happen on the top layer. Hence, the number N of molecules on the grain entirely determines the geometry of the ice mantle : every layer starting from the bare grain's surface is filled with N_s molecules each, with the top one being possibly incomplete.

To express the size of ice mantles, we will often give the *number of monolayers* N/N_s rather than the number of molecules N, as it is more representative of the thickness of the ice.

Grain surface stochastic models usually set an upper limit on population ([31, 32, 38]) (which can help in the mathematical resolution by simplifying infinite systems to finite ones). As we are interested in the possibility for accumulation of species and the formation of ice mantles, we do not set such a limit and let the population N be arbitrarly large.

We now turn to the prescription of the rates at which the adsorption process and the desorption processes take place.

2.1.4. *Adsorption*

The adsorption of species happens through the collision of gas phase particles with the grain. The frequency at which the grain collides with molecules in the gas depends on the mean speed of considered molecules in the gaz $\langle|v(\mathrm{A})|\rangle$, the density of molecules in the gas $n_{gas}(X)$, and the geometric cross section πa^2 of the grain. The collision rate (in s^{-1}) thus writes:

$$k_{coll} = \pi a^2 \langle|v(\mathrm{A})|\rangle n_{gas}(\mathrm{A}).$$

We take the velocity to be the mean thermal velocity of particles in a Maxwellian distribution

$$\langle|v(\mathrm{A})|\rangle = \sqrt{\frac{8}{\pi}\frac{kT_{gas}}{m_\mathrm{A}}},$$

where T_{gas} denotes the gas temperature and m_A is the mass of an A molecule. Contrary to the case of atomic hydrogen, for which a sticking probability is usually introduced, our molecules of X are heavier and always stick to the grain after collision. In this setting, the effective adsorption rate is exactly the collision rate $k_{ads} = k_{coll}$. In a rate equation formulation, the population increase per second due to adsorption is

$$\left.\frac{dN}{dt}\right|_{ads} = k_{ads} = n_{gas}(\mathrm{A})\sqrt{T_{gas}}a^2\sqrt{\frac{8\pi k}{m_\mathrm{A}}}.$$

Note that it only depends on the gas conditions (density and temperature), moreover those conditions appear as degenerate parameters since the true dependence is only on the product $n_{gas}(\mathrm{A})\sqrt{T_{gas}}$.

2.1.5. *Desorption*

Molecules can desorb by thermal evaporation, a central process in our analysis since its rate is controlled by grain temperature. Since we are interested in the formation and destruction of ice mantles in UV-illuminated regions, such as edges of clouds and disks, we also consider photodesorption. Other non-thermal desorption processes are expected to be negligible in our setting.

Thermal desorption. The adsorption site is modelled by a potential well of depth kT_{bind}, from which particles can escape simply by thermal vibration, provided the grain is hot enough. The desorption probability per second and per occupied site is modeled as usual [27, 22] with an exponential dependence on grain temperature:

$$k_{th} = \nu_0 \exp\left(-\frac{T_{bind}}{T_d}\right).$$

The binding energies kT_{bind} vary between species and condition its volatility. Values for CO and H_2O are given in table 1. The quantity ν_0 is a typical vibration frequency in the site:

$$\nu_0 = \sqrt{\frac{2kT_{bind}}{\pi^2 d_0^2 m_{\mathrm{A}}}}$$

(where $d_0 = 0.1\mathrm{nm}$ is the typical width of the potential well). However, recent experimental measurements of thermal desorption find ν_0 values that differ from this analytical formula. We will prefer these recent recommended values when they are available (for instance those of [21]).

The rate k_{th} is the desorption rate per occupied site, and should then be multiplied by the number N of currently occupied sites to obtain the total rate for the grain:

$$\left.\frac{dN}{dt}\right|_{th} = -Nk_{th}(T_d).$$

We shall remark that this formula is not entirely correct: as recommended in the review by Cuppen et al. [26] and as it is done in some models (for instance Grainoble [39]), one should take into account that molecules trapped deep in the mantle cannot desorb as easily as ones on the top layers. This implies that one should fix a number l_{act} of such desorption-active layers from which molecules can desorb, and that, if there are more than l_{act} layers, only the $l_{act}N_s$ molecules from those layers can desorb. This amounts to replacing the N term in the above equation by the bound $\min(N, l_{act}N_s)$. As we are interested in the formation of ice mantles at the edge of clouds, we expect to be in a low-population regime with only a few layers, and we will ignore this correction.

Photodesorption. Photodesorption designates the desorption of molecules due to a photon directly breaking the bond between a molecule and the mantle. Laboratory experiments report that it can involve complex and hard to model mechanisms, which in models are all conveniently encompassed in a single resulting rate of mean number of molecules desorbed per incident photon, known as the yield Y. However, careful considerations are needed to correctly interpret the experimental data. As they are not central to our discussion, we postpone them to the Appendix (see A). The final formula that we implement is the following:

$$\left.\frac{dN}{dt}\right|_{ph} = -\frac{Y 4\pi^2 a^2}{l_{act}N_s}\left(\int_{UV}\frac{I_U(U)}{U}dU\right)N = -Nk_{ph}.$$

We see that the rate per occupied site k_{ph} depends on the number of UV photons hitting the grain, through the flux of UV photons (integral term) and the grain cross section. The yield Y is given by experimental data (the values used are in Table 1). The $l_{act}N_s$ factor results from the considerations done in appendix. The rate is multiplied by the number of molecules N available for desorption.

	Yield Y	T_{bind} [K]	ν_0 [s^{-1}]	d_s [nm]
H_2O	$4 \cdot 10^{-4}$	5705	$4.96 \cdot 10^{15}$	0.26
CO	10^{-3}	1390	$9.14 \cdot 10^{14}$	0.26

Table 1: Values of the parameters used for H_2O and CO ices. Thermal desorption parameters are taken from [21] and yields from [40], distance between sites from [27]

Photodesorption and thermal evaporation are both linear in N and the two rates can be combined in a total desorption rate $k_{des}(T_d) = k_{ph} + k_{th}(T_d)$. This rate is constant at the low temperatures (dominated by photodesorption), and is exponential at higher temperatures (dominated by thermal desorption). The switching point between those two regimes is the temperature for which the contributions from both mechanisms are the same, i.e. $k_{ph} = k_{th}(T_d^{switch})$. The values for CO and H_2O are respectively $T_d^{switch} = 23K$ and $T_d^{switch} = 92K$. From those values, one can already foresee that temperature fluctuations are more likely to trigger thermal desorption spikes of CO because it is much more volatile, while grains must reach nearly a hundred Kelvin to thermally desorb H_2O efficiently.

These values were computed using a photodesorption rate k_{ph} in standard ISRF conditions, that is without extinction ($A_V = 0$) nor scaling of the UV part of the field ($\chi = 1$). However, the dependence of the switching temperature is mainly on the binding energies of the species, as one can see from the full normalized formulas that we write here for the sake of completeness:

$$\begin{aligned} T_d^{switch} &= 23 \left(\frac{T_{bind}}{1390K}\right) \left[\ln\left(\frac{\nu_0}{9.14 \cdot 10^{14}\mathrm{s}^{-1}} \frac{1.7 \cdot 10^{-11}\mathrm{s}^{-1}}{k_{ph}}\right)\right]^{-1} K \\ &= 92 \left(\frac{T_{bind}}{5705K}\right) \left[\ln\left(\frac{\nu_0}{4.96 \cdot 10^{15}\mathrm{s}^{-1}} \frac{6.7 \cdot 10^{-12}\mathrm{s}^{-1}}{k_{ph}}\right)\right]^{-1} K. \end{aligned}$$

We have now covered all the relevent processes at play in the formation of ice mantles: the absorption and emission of photons controlling the temperature T_d, and the adsorption and desorption of species controlling the population on the grain N and the size of the ice mantle.

2.1.6. *Rate equation solution*

The rate equation for the adsorption and desorption of species writes:

$$\frac{dN}{dt} = k_{ads} - N k_{des}(T) \tag{6}$$

In the absence of temperature fluctuations, T is constant, fixed to the equilibrium temperature T_d^{eq} given by equation (5). The steady state solution of (6) is the equilibrium population

$$N^{eq}(T_d^{eq}) = \frac{k_{ads}}{k_{des}} = \frac{k_{ads}}{k_{ph} + \nu_0 e^{-T_{bind}/T_d^{eq}}} \tag{7}$$

which is independent of temperature if $T_d^{eq} < T_d^{switch}$. Indeed, in this regime photodesorption regulates the growth of the ice mantle. On the opposite it decays exponentially with the equilibrium temperature in the other regime $T_d^{eq} > T_d^{switch}$. For H_2O, this latter regime requires extreme conditions since it would imply that the grains are hotter than 92K at equilibrium. However, for CO, the proximity of a bright star can easily lead to grain temperature higher than 23K at the edge of clouds.

Note that N^{eq} is proportional to k_{ads}, so simply scales with the gas conditions $N^{eq} \propto n_{gas}(\mathrm{A})\sqrt{T_{gas}}$. It will also be the case of the solution to the stochastic problem. This will prove convenient because it implies that the relative differences between the rate solution and the stochastic solution are independent of the gas conditions.

Now that the rate solution of the ice mantle size given by (7) provides us with a comparison point, we turn to the master equation approach to assess the influence of stochastic fluctuations.

2.2. Master equation approach

2.2.1. *Stochastic formulation*

In the stochastic approach, the thermal energy $E(t)$ (and hence the grain temperature T_d) and the surface population $N(t)$ are modelled as coupled stochastic processes. To make use of the abstract general formulation presented in introduction, we define the states X to be the data of a possible thermal energy and population of the grain, $X = (E,N) \in \mathbb{R}^+ \times \mathbb{N}$. The probability distribution function f of the states $X = (E,N)$ in the stationary regime must obey the Master Equation:

$$\int_{\mathbb{R}^+\times\mathbb{N}} f(Y) p_{Y\to X} dY = f(X) \int_{\mathbb{R}^+\times\mathbb{N}} p_{X\to Y} dY. \tag{8}$$

Here the integral over $\mathbb{N}$ should be understood as discrete summation. The transition rates $p_{X\to Y}$ represent the different possible transitions of the system from a state $X = (E,N)$ to a different state $Y = (E',N')$. These transitions should correspond to the physical processes described above: grains receive and emit photons, as well as adsorb and desorb molecules and atoms. Each of these physical events is supposed to occur as a Poisson process. The allowed transitions should then be:

- Transitions from E to any $E' > E$, at fixed N, corresponding to the absorption of a photon of energy $E' - E$. This happens at a rate $R_{abs}(E' - E)$.
- Transitions from E to any positive $E' < E$, at fixed N, corresponding to the emission of a photon of energy $E - E'$. This happens at a rate $R_{em}(E - E')$.
- Transitions from N to $N' = N + 1$, at fixed E, corresponding to adsorption. This happens at a rate k_{ads}.
- Transitions from N to $N' = N - 1$, at fixed E, corresponding to desorption (thermal or photodesorption). This happens at a rate $k_{des}(E)$.

Remark that there is no transition which changes both energy and population at the same time. We can thus write $p_{X\to Y}$ as the sum of energy transitions at fixed N and population transitions at fixed E, making use of the Dirac δ function:

$$p_{X\to Y} = p_{E\to E'}\delta_{N=N'} + p_{N\to N'}\delta_{E=E'}$$

Moreover, we can explicit these two transition rates as:

$$p_{E\to E'} = \begin{cases} R_{abs}(E' - E) & \text{if } E' > E \\ R_{em}(E - E', E) & \text{if } E' < E \end{cases}$$

and

$$p_{N\to N'} = \begin{cases} k_{ads} & \text{if } N' = N + 1 \\ k_{des}(E)N & \text{if } N' = N - 1 \\ 0 & \text{otherwise} \end{cases}$$

At this point we could explicit the master equation for $X = (E,N)$. However, because the problem is two dimensional, it is very costly to numerically solve as is: we should discretize the 2D plane of possible energies and populations (the latter are already discrete, but with too much possible values to consider them all), but a 2D discretization grid quickly contains an overwhelming number of points. Instead, we can do some analytical work and first derive an equation for the thermal energy alone, and then an equation for the population knowing the temperature. This reduces the 2D problem to two 1D ones.

2.2.2. *Energy and temperature PDF*

The population does not affect the heating and cooling of the grain, so the thermal energy E should not be coupled to the population N. This translates in the fact that the energy transition rates do not depend on N, which means that the marginal PDF for the energy, $f(E)$, can be independently solved, which simplifies the resolution (This is analogous to the fact that we could solve for the equilibrium temperature independently of the grain population in the previous section). The PDF $f(E)$ is obtained by the law of total probability as $f(E) = \sum_{N=0}^{\infty} f(E,N)$. One sums equation (8) over N to obtain:

$$\int_{\mathbb{R}^+} f(E')p_{E'\to E}dE' = f(E)\int_{\mathbb{R}^+} p_{E\to E'}dE'. \tag{9}$$

This is in fact the Master Equation expressed for the stochastic process E only. It can be recast as a Fredholm equation of the second kind by defining the linear operator $\mathcal{L}$:

$$\mathcal{L}[g](E) = \int_{\mathbb{R}^+} g(E')\frac{p_{E'\to E}}{\int_{\mathbb{R}^+} p_{E\to F}dF}dE'.$$

In these terms, Eq. (9) becomes an eigenproblem $\mathcal{L}[f] = f$, which is solved numerically (see section 2.3). As the thermal energy and the temperature are ono-to-one related quantities, the thermal energy PDF $f(E)$ can be converted to the temperature PDF $f_T(T_d)$, using the relation $f(E)dE = f_T(T_d)dT$, which gives $f_T(T_d) = f(E)C(T_d)$.

2.2.3. *Conditional mean surface population*

We now suppose that we know the thermal energy distribution $f(E)$ thanks to equation (9). This allows us to avoid the computation of the full distribution $f(E,N)$. Indeed, we want to know in the end the typical size of ice mantles under fluctuations, so we can satisfy ourselves with knowing only the mean population on the grains, $\langle N\rangle = \int Nf(E,N)dEdN$. This mean value is an average population that takes into account all stochastic fluctuations, and this is the quantity that we wish to compare to the rate equation solution N^{eq}. In the computation of $\langle N\rangle$, an intermediary quantity is the mean population conditioned on E, $\langle N|E\rangle$. This quantity is defined as:

$$\langle N|E\rangle = \frac{\sum_{N=0}^{\infty} Nf(N,E)}{f(E)},$$

and is the mean population one can expect when observing only grains at a temperature E. One can deduce the mean population by the relation

$$\langle N\rangle = \int_{\mathbb{R}^+} \langle N|E\rangle f(E)dE.$$

Adapting the method used in [1], one can obtain an eigenvector equation for the conditional mean population, similar to the equation (9) for the thermal PDF. The derivation is given in Appendix F.1. The final formulation is:

$$\langle N|E\rangle = \int_{\mathbb{R}^+} \langle N|E'\rangle\kappa(E,E')dE' \tag{10}$$

where the kernel κ is given by:

$$\kappa(E,E') = \frac{f(E')}{f(E)}\frac{p_{E'\to E} + f(E)k_{des}(E')}{k_{des}(E) + \int_{\mathbb{R}^+} p_{E\to F}dF}.$$

The equation (10) being homogeneous, we also derive the normalization condition:

$$\int_{\mathbb{R}^+} \langle N|E\rangle f(E)k_{des}dE = k_{ads} \tag{11}$$

As k_{ads} only appears in the normalization and not the kernel, the solution $\langle N|E\rangle$ is directly proportional to k_{ads}, and hence proportional to the gas condition parameter $n_{gas}(X)\sqrt{T_{gas}}$, as it was the

case for the rate equation solution. This means that a solution $\langle N|E\rangle$ holds for the full range of gas conditions, provided a suitable rescaling. Equation (10) is once again an eigenvector equation for the eigenvalue one of a different integral operator. We have thus reduced the full two-dimensional master equation (8) to the two eigenproblems (9) and (10) for functions of one variable only. The numerical methods to find first $f(E)$ and then $\langle N|E\rangle$ are briefly detailed in the next section, but most technical elements of the implementation are postponed to the appendices.

2.3. Numerical resolution

The presentation of the numerical method will be kept extremely short and as computation-free as possible. What we need to solve is an eigenproblem in a space of functions, of the form $\mathcal{L}[f] = f$. The operator $\mathcal{L}$ is of the form:

$$\mathcal{L}[f](E) = \int_{\mathbb{R}^+} f(E')\kappa(E,E')dE' \tag{12}$$

The strategy is as follows: we first build a discretization of the operator $\mathcal{L}$ itself on a finite-dimensional subspace of the function space, which allows us to write it as a matrix L. We then solve the equivalent eigenproblem for the matrix L: we find its eigenvector for the eigenvalue 1 (or closest to 1 because of numerical errors). Because it is now a finite dimensional eigenproblem, we can rely on the whole panel of numerical linear algebra techniques.

The first step in the discretization of $\mathcal{L}$ is to transform the integral in its definition into a finite sum. Because the incoming photons that heat the grains have their energy under the Lyman limit, and that the grain's cooling efficiency increases as it heats up, there should be a maximal energy E_{max} that the grain has virtually no chance of exceeding. We can thus restrain the integral to the finite energy domain $[0,E_{max}]$. We then introduce a discretization grid of this interval $0 = E_0, E_1, ..., E_I = E_{max}$ and we only consider piece-wise affine functions g on this grid. This is a finite-dimensional space of functions, which are entirely determined by their values $g_i = g(E_i)$. We define the matrix L as the matrix representation of the finite-dimensional operator that, to a piece-wise affine function g given by the $(g_i)_i$, associates the piece-wise affine function given by the values $(\mathcal{L}[g](E_i))_i$. Details on the grid and the value of the coefficients of the matrix L are given in Appendix G.1.

To solve for the eigenvector of value 1 of L, we rely on an iterative method. Indeed, one can show that the eigenspace for 1 is one-dimensional (meaning only one function is solution up to a multiplicative constant), and all the other eigenvalues of $\mathcal{L}$ are in absolute value smaller than 1. This means that iterating L on a non-pathological initial guess converges to the solution, as the components on the other eigenspaces vanish to 0. A straightforward diagonalization of L is also a possible method, it yields reasonable results but remains less precise than the iterative method. We refer to the Appendix (G.2) for details on the iterative scheme. This numerical strategy was implemented in a standalone Fortran code named `StoCapy`. We now turn to the results of the stochastic model.

2.4. Results

We solved our stochastic model for grain sizes a ranging from $\sim$ 1nm to $\sim$ 300nm. We first ran a model with standard ISRF, meaning we did not scale the UV part of the field ($\chi = 1$) and supposed there was no extinction (as if we considered the very edge of a cloud, $A_V = 0$). This allows us to concentrate on the stochastic effects on grain temperature and population in a typical setting of the ISM. The grain temperature PDF in these conditions had already been derived in [31] when the formation of H_2 was studied. Results on the ice mantles, however, are new since this is precisely the reason for this study.

We also ran two grids of models with higher UV-illumination. For the first one, we scaled the UV field to $\chi = 100$ to simulate a medium PDR such as the Horsehead Nebula. For the second one, we set $\chi = 10^4$ to model the intense UV-illumination of a PDR like the Orion bar. Those two regions are often studied as prototypes of medium and strong PDRs. For both, we modelled the depth into the cloud by letting A_V range from 0 to 20. The impact of extinction had not been covered before in the literature and, as we will see, present interesting results.

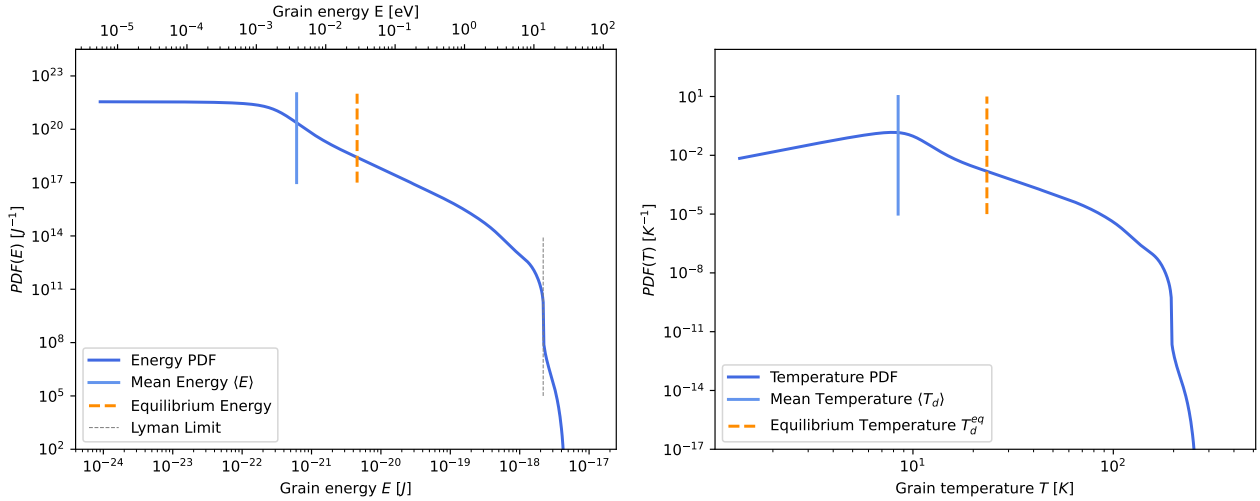


Figure 4: Energy (left) and Temperature (right) PDF of a small $a = 2\text{nm}$ grain (under standard ISRF, $\chi = 1$, $A_V = 0$).

As explained above, we considered CO or H_2O as species constituting the grain mantle, in order to highlight the differences between a more volatile and a more refractory mantle. We begin by investigating the distribution of grain temperatures alone, before covering its effects on the ice mantles.

2.4.1. *Grain temperature*

Temperature under standard ISRF. We first discuss the grain temperature under standard ISRF conditions ($\chi = 1$ $A_V = 0$). We recall that the usual determination of temperature is obtained by supposing thermal equilibrium and solving equation (5). It amounts to consider that the grain receives heat through a continuous constant-over-time flux, rather than individual photons ; and that it emits a continuous power as well. This will be compared with the full distribution of possible thermal energies $f(E)$ and —equivalently— of temperatures $f_T(T_d)$, provided by our stochastic model. We will also consider the associated mean temperature:

$$\langle T_d \rangle = \int T_d f_T(T_d) dT_d.$$

We begin our investigation with the smaller grains. Indeed, they are the ones with the smallest heat capacity, so their temperature is subject to the widest fluctuations upon absorption of a photon. The PDF of the thermal energy f and of the temperature f_T of a 2nm grain are showcased in Figure 4, along with the mean values of the distributions, and the equilibrium values obtained by the rate treatment (the solution T_d^{eq} to equation (5)). The first striking result is the spread of the distribution. It extends from a non zero limit at zero energy, up to a sharp cut-off at the Lyman limit ($2.18 \cdot 10^{-18} J$ or 13.6eV). The wideness of the PDF expresses how far the grain is from achieving thermal equilibrium: it can be found with an appreciable probability in a large range of temperatures, from lower than T_{CMB} =2.73K, to nearly 200K. Its bulk is located below equilibrium temperature, but its tail extends up to several hundred Kelvin.

The different regions of the PDF correspond to different moments in the grain's cycle of temperature spikes. The majority of the PDF is supported on low temperature (below the mean one): those are the cold states at which the grain rests between two energetic photon absorptions. The non-zero limit at $E = 0$ means that the ambient radiation field does not guarantee a minimal temperature of the grain, and it is not unlikely for the grain to emits almost all of its energy in one photon. We stress that this differs from a standard treatment of cooling: if the grain emitted a continuous power as a modified Stefan's law $P_{em} \propto T^\beta$, it would take an infinite time to reach zero energy (and hence would never happen because the grain would eventually heat up again). The long tail from 10^{-21}J

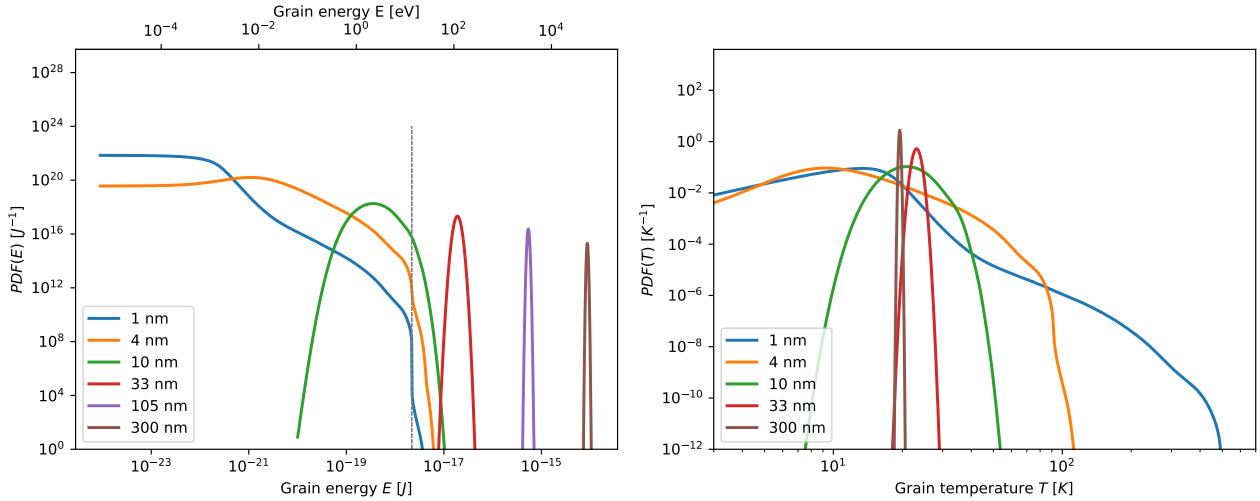


Figure 5: Energy (left) and Temperature (right) PDFs for various grain sizes. The dashed grey line is the Lyman limit ($2.18 \cdot 10^{-18}$J / 13.6eV). The grain $a = 105\text{nm}$ was omitted from the right panel for better readability.

to $2 \cdot 10^{-18}$J (15K to 200K), corresponds to photons cooling down from a temperature spike after absorption of a UV photon. The negative slope is the combination of the states becoming harder to reach (because it requires increasingly energetic photons), and the grain spending less time at those higher energies since it cools down faster at higher temperature (it has a higher emitted power). The Lyman cut-off is because no photon of higher energy is present in the radiation field (see Section 2.1.2). Hence, grains above the Lyman limit must have absorbed two UV photons in a row before any significant cooling took place, which is quite unlikely. A second cut-off at two times the Lyman limit further attests of this behaviour.

Another striking effect is the difference between mean and equilibrium temperature. We wish to stress that not only is the distribution too wide to be reduced to its mean value, but the mean value itself is not even close to the equilibrium one. The mean temperature $\langle T \rangle = 9K$ is actually far colder than the equilibrium temperature $T^{eq} = 24K$. In fact, the two have no reason to match up, because the mean temperature is a balance between the long times spent at low energies and the short temperature spikes and their subsequent cooling phase. This balance has no reason to correspond to the equilibrium temperature. The mean temperature is dependent on the temperature spikes, which are the consequence of photons of energies higher than the grain's. On the other hand equilibrium temperature does not take into account from which spectral domain the absorbed power comes: the power could be coming in many microwave photons rather than a single UV photon and the equilibrium temperature would be the same. However, in this setting, small photons would not entail temperature spikes and the mean temperature would be close to the equilibrium one. The wideness of the PDF and its shift compared to equilibrium are both expected to have an effect on any temperature-dependent process.

We now study the evolution of the PDF with the size of the grain. Figure 5 displays energy and temperature PDFs for different grain radii. As the grain grows, so does its heat capacity, and thus its thermal energy (recall that $E(T_d) \propto a^3$). For larger grains, the absorption of a single UV photon entails a smaller relative change in energy and a smaller temperature spike. The spread of the temperature PDF gets narrower for larger grain sizes. In the limit where the typical energy is greater than that of a Lyman photon, $E \gg U_{Lyman}$, the grain never undergoes more than small oscillations around its equilibrium temperature, and its PDF spikes around this value. Both of the effects discussed above disappear: the mean and equilibrium temperatures start to match, and the PDF sharpens around this temperature.

The sharpening of the PDF and the reconciliation of equilibrium and mean temperature is further

identified in Figure 6. It showcases equilibrium and mean temperatures in function of size, in order to identify the range of sizes affected by the stochastic fluctuations. While the equilibrium temperature plateaus for low to medium grain sizes, the mean temperature dips as the stochastic nature of photon absorption becomes relevant. The dotted lines plot the quantity $\langle T\rangle \pm \sigma$ where σ is the usual standard deviation $\sigma^2 = \langle |T - \langle T\rangle|^2\rangle$. This indicator is not the most suited to our situation since the PDF is highly asymmetric, yet it still allows one to visualize easily the narrowing of the PDF as the grain grows. As can be inferred from Figure 6, for grains larger than $a \gtrsim 30\text{nm}$ the stochastic effects vanish and we expect the rate treatment to yield correct results when modelling temperature-dependent processes. This size corresponds to the grain thermal energy at equilibrium becoming comparable to the energy of a Lyman photon.

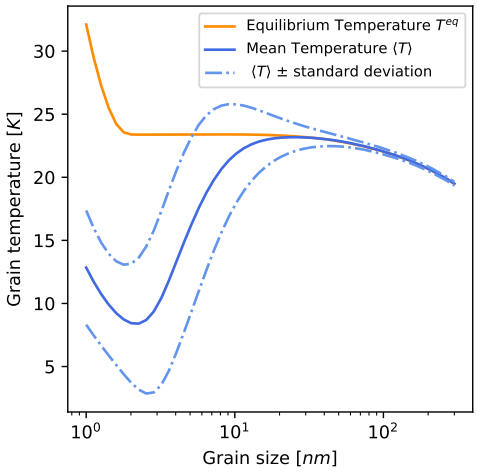


Figure 6: Equilibrium and mean grain temperatures vs. grain size.

The rise in temperature under 2nm is due to the impossibility of the grain to emit photons of energies greater than its own thermal energy, modelled mathematically by the upper bound $E(T_d)$ (rather than ∞) in the integral defining the emitted power P_{em} in equation (5). This is not a stochastic effect as it affects both equilibrium and mean temperature. It effectively cuts down on the very small grains' cooling abilities.

Temperature in function of radiation field conditions A_V and χ. We now present the results of the two grids of models that we ran, to study the influence of the radiation field through the extinction A_V and the scaling factor χ. The first grid simulated a cloud edge with a medium UV flux of $\chi = 100$ such as the Horsehead Nebula, and the second grid had $\chi = 10^4$ to model a strongly illuminated region like the Orion bar. As said above, these environments are two prototypes of PDRs. We ran both grids for extinctions $0 \leq A_V \leq 20$ to model the depth into the cloud, and the usual grain sizes $1\text{nm} \leq a \leq 300\text{nm}$.

The grain mean and equilibrium temperatures in function of grain size are shown in Figure 7. The major effects are as expected: the $\chi = 10^4$ environment presents generally higher grain temperatures

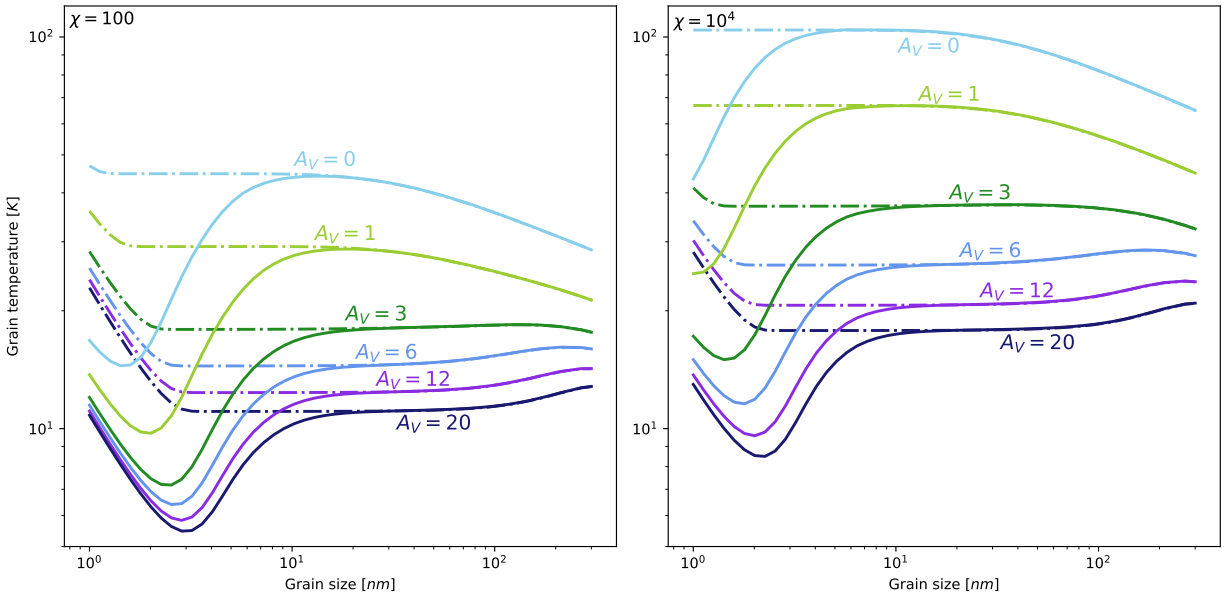


Figure 7: Mean temperature (plain) and equilibrium temperature (dotted) in function of grain size, for different extinctions A_V. Left panel is $\chi = 100$, right panel $\chi = 10^4$.

than $\chi = 100$ because of the stronger influx of UV photons. The progressive extinction of the field lowers the grain temperatures from 40K to around 10K (for $\chi = 100$) and from 100K to 15K (for $\chi = 10^4$) as we go from $A_V = 0$ to 20.

However, some additional effects deserve some discussion. On the scaling factor χ, one can see that at $A_V = 0$, the grain sizes range affected by stochastic effects is smaller for higher χ (for $\chi = 100$, grains as large as 10nm are affected, while for $\chi = 10^4$, only grains up to 5 nm are affected). This could be counter-intuitive, as more frequent UV photons would make one expect more fluctuations and stochastic effects. It reflects that, in fact, since grains are at a higher typical temperature because of the stronger radiation flux, the size such that $E \simeq U_{Lyman}$ (under which thermal equilibrium becomes impossible) gets smaller the higher the value of χ.

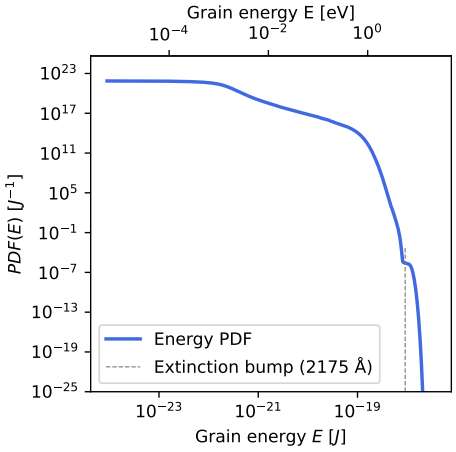


Figure 8: Energy PDF of a 2nm grain under radiation field $\chi = 100$, $A_V = 20$.

Another result worth discussion is this time at high A_V: one could expect the temperature fluctuations to progressively disappear as A_V increases, since the extinction suppresses nearly all UV photons, which were causing the temperature spikes of small grains. Nonetheless, stochastic effects remain very much visible, for a range of sizes that is roughly independent of extinction. While it is true that UV photons disappear, the fact that grains get colder counterbalances this and makes a stochastic treatment relevant even in the depths of molecular clouds. Indeed, the extinction of UV photons also entails that the grains are less heated. As they have a lower typical thermal energy, they are more easily kept out of equilibrium by the absorption of a photon. When nearly all UV photons are extinguished, the grains' thermal energies have dropped so low that even photons in the visible and near IR spectra can trigger noticeable temperature spikes.

This is best illustrated on Figure 8 showcasing the PDF of a grain at $A_V = 20$. The grain actually very rarely reaches the Lyman limit ($2.18 \cdot 10^{-18} J$), and the cut-off of the PDF is at $9 \cdot 10^{-19} J$. This corresponds to the extinction bump at 2175 $\dot{A}$ [37] (see Figure 3 in Section 2.1.2). It produces a drop in the radiation field which plays the same role at high A_V as the Lyman cut-off at lower A_V. To our knowledge, light had never been shed on this effect, that makes the stochastic treatment of grain temperature relevent in a wider range of conditions than expected.

Summary. From the stochastic treatment of temperature, two main effects are highlighted. First, the temperature of small grains follows a very wide distribution (reaching from 0K to above 100K), which cannot be reduced to a single temperature as assumed in classical astrochemistry models. This distribution has its bulk at colder temperatures than equilibrium (by 15K under standard ISRF, and increasing to tens of Kelvin in strongly illuminated regions). The grain spends most of their time in this bulk, but the distribution also extends way above equilibrium. The stochastic effects can be important for grains as big as 30nm.

Second, temperature fluctuations are still relevant in the depth of clouds (under an extinct radiation field) because the colder grains are more easily sent out of equilibrium by visible and near-IR photons. This effect had never been discussed previously in the literature. We now study the impact of these temperature fluctuations on ice mantles.

2.4.2. Surface population under standard ISRF

For the surface population, we recall that CO and H_2O were considered, as they are two of the main constituents of interstellar ice, and provide respectively a volatile and refractory species. We used the values from Table 1 for the model parameters such as binding energies and photodesorption yields. In this section we discuss the population under standard ISRF conditions (as usual, this means $\chi = 1$, $A_V = 0$). The two grids simulating UV-illuminated regions will be discussed in the subsequent section.

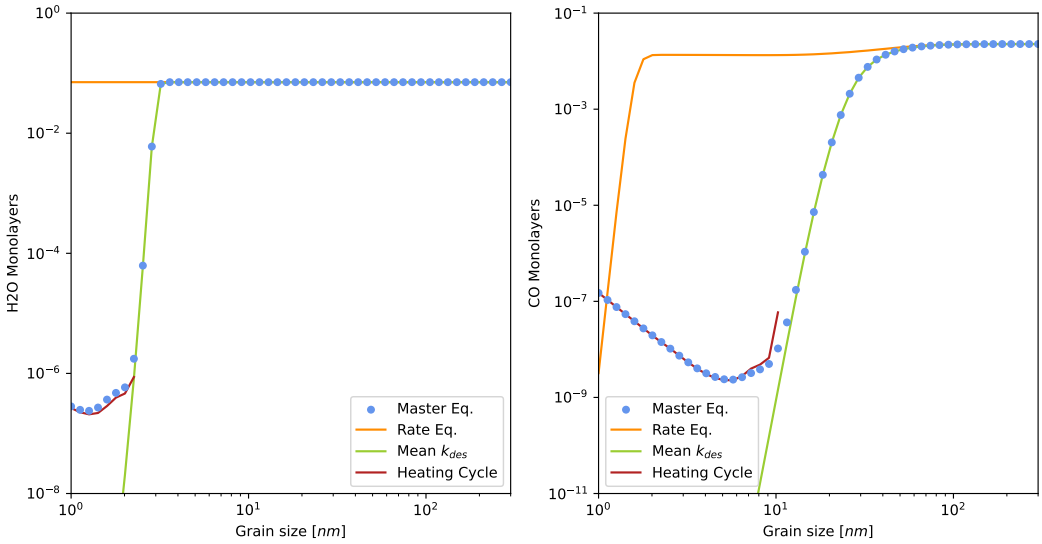


Figure 9: Ice mantle population (in monolayers) in function of grain size, for H_2O (left) and CO (right). The mean population from the master equation, the rate equation population, and the Mean k_{des} and Heating Cycle approximations are plotted. Under standard ISRF, $\chi = 1$, $A_V = 0$.

Before diving into the results, we discuss the choice and effect of gas conditions (temperature and density). In our models, the gas conditions were kept fixed to $n_{gas}(\mathrm{A})\sqrt{T_{gas}} = 1\mathrm{cm}^{-3}K^{1/2}$, for both A=CO and A=$H_2O$. For a gas temperature of 15K and a proton density of $n_H = 10^4\mathrm{cm}^{-3}$, this corresponds to an abundance of A of $n_{gas}(\mathrm{A})/n_H = 2.6 \cdot 10^{-5}$. However, the true abundance of A in a cloud is coupled with the gas-phase chemical reactions involving A, but also with the size of the ice mantles, as their formation progressively depletes the gas phase. One should compute the full gas-grain chemistry equilibrium to obtain the exact values of densities of A, both in the gas and adsorbed on dust. This equilibrium depends on the amount of grains and their size distribution, and couples all the grain sizes together. Nonetheless, we recall that the results of both the rate and the master equations are both simply proportional to the gas conditions $n_{gas}(\mathrm{A})\sqrt{T_{gas}}$. We do not solve for the exact value of $n_{gas}(\mathrm{A})$, since it would require to run a full chemistry model, but it would not affect the shape of the results. We rather assume that the gas phase reservoir is constant. Because both results scale the same way, this does not affect the difference between the rate equation and the master equation. Of course, it would be interesting to run two full chemistry simulations to frame the differences between the two approaches in a larger picture, but this is out of the scope of this project.

General behaviour. We begin by investigating the rate equation solutions. These are plotted on Fig. 9 as orange lines. We recall that we will work with the number of monolayers N/N_s, which is a more common quantification of the size of the mantles, rather than the sole population N. For H_2O, the number of monolayers is independent of grain size: this can be explained by two effects. First, the equilibrium temperature of grains ($\sim 20K$) is way under the temperature at which thermal desorption becomes efficient ($T^{switch} = 92K$ in these conditions), so the population is entirely regulated by photodesorption. The exact value of the equilibrium temperature —which is grain-size dependent, recall Figure 6— hence does not play a role because thermal evaporation is negligible. The only dependence on grain size of the population N is through the collision rate $k_{ads} \propto a^2$, but so is the number of sites per layer $N_s \propto a^2$. Every grain thus has the same number of ice layers. For CO, on the contrary, the switching temperature is much lower: T^{switch} =23K. The smaller grains have a higher equilibrium temperature than 23K, so their population is regulated by thermal desorption (why their temperature is higher than bigger grains is because their low energy cuts down on their cooling ability, as discussed previously). This explains the drop under 2nm. However, apart for these very small grains, with a rate formalism and the use of equilibrium temperature, we would deduce

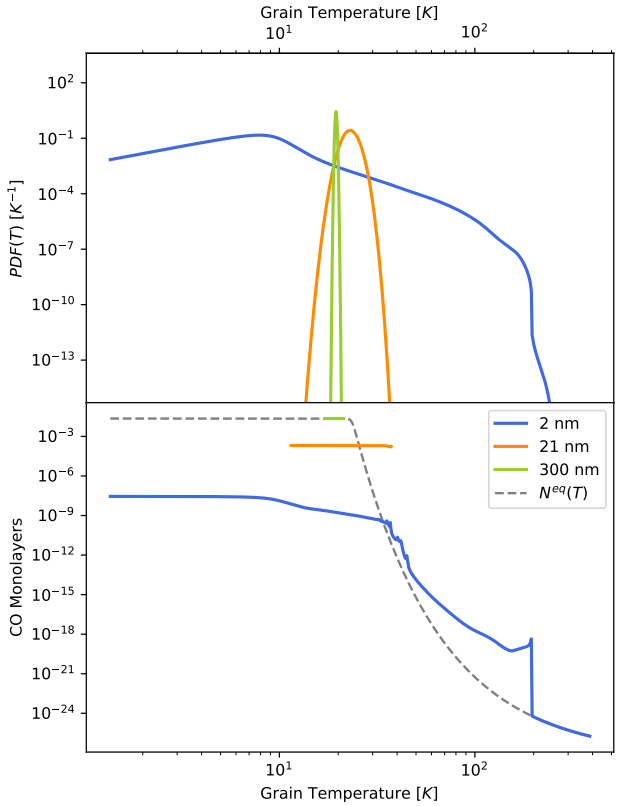


Figure 10: *Top:* Temperature PDF of three grain sizes in each of the regimes for CO (see text). *Bottom:* Conditional mean population $\langle N|T\rangle$ for the same sizes (plain lines, expressed in layers), and rate equation solution (dashed). Under standard ISRF, $\chi = 1$, $A_V = 0$.

that the thermal desorption never plays a role on the number of H_2O or CO monolayers.

The stochastic computations exhibits that the temperature fluctuations, ignored by the equilibrium approach, do impact the smaller grain sizes as seen on Figure 9 (Master Eq. blue dots). If the affected range varies between species, in both cases three regimes can be identified. The first one is that of the small grains, which have a much lower mean population than equilibrium (sizes up to 2nm for H_2O and up to 10nm for CO). The second one is the intermediate sizes (between 2 and 3nm for H_2O and 10 to 40nm for CO), for which the mean population catches up with the equilibrium computation as the grain grows. The third regime is that of big grains whose population is unaffected by temperature fluctuations. The population on small grains is overestimated by several orders of magnitude by the equilibrium-temperature rate-equation approach. This affects only very small grains for H_2O ($< 3\text{nm}$), but in the case of CO a significant part of the distribution has its population greatly overestimated (sizes $a < 20\text{nm}$). Recalling that small grains are the most abundant in the ISM, the quantity of CO ice is all the more overestimated by the rate method.

We will now investigate each of the three regimes. Our understanding of the mechanisms at play in each one will also allow us to build approximations of the full result. Figure 10 displays the temperature PDF and conditional mean population $\langle N|T\rangle$ (in monolayers) for the three regimes.

Small grain regime. The first regime is illustrated by a 2nm grain on Fig. 10. The main highlight is that the population plateaus at low temperatures not at the rate equation value but at a much lower one (by 6 orders of magnitude). In both cases less than a monolayer is formed. The difference between the two approaches is because UV photon heating events periodically bring the grain to high temperatures at which thermal desorption is very effective. This can be seen from the position of the tail of the PDF: the grain has an appreciable probability of being brought to such temperatures. Such heating events happen sufficiently often to prevent the grain from reaching its full equilibrium population at low temperature.

After the plateaus ends, between 40K and 200K, we observe that the grain population is higher than its equilibrium value at the same temperature: grains that absorb a photon and are raised to those high temperatures bring their low-temperature population with them, which does not have the time to fully desorb before the grain cools down.

Above 200K, the mean population matches with the equilibrium one, and this is the combination of two effects. First, the timescale for the population desorbing and reaching equilibrium is $1/k_{des}(T_d)$, which gets exponentially faster with temperature. The desorption rate increases faster than the rate at which the grain cools down, so at very high temperature the desorption happens very quickly and the population reaches equilibrium before any significant cooling took place. The second effect is that the grain being above 200K implies that it absorbed two UV photons in a row: it is beyond the Lyman limit cut, as shown on the PDF. The few grains that reach this state do not arrive

with the high population of the plateau but the already partially desorbed one from the intermediary temperature range.

Heating Cycle Approximation for the small grain regime. We can use the physical explanations above to build an approximation of the mean population in this regime. Since small grains are periodically nearly emptied by the heating events, the value of the low population plateau corresponds to the average population that the grain has the time to adsorb before absorbing a UV photon. This allows us to build an approximation of $\langle N \rangle$ (which is very close to the value of the plateau as the bulk of the PDF is over those temperatures). Indeed, if the mean time between two photon adsorptions is t_{heat}, the population that the grain can accumulate is $N = t_{heat} k_{ads}$. The time t_{heat} is the inverse of the rate at which the grain is brought to high temperatures from a very low one, that is:

$$\frac{1}{t_{heat}} = \int_{E_{crit}}^{\infty} R_{abs}(E')dE'. \tag{13}$$

In this expression, we have designated high energies to be the ones above a certain threshold E_{crit}. To reach $E' > E_{crit}$ from an energy E, the rate would be in all generality $R_{abs}(E' - E)$, but $E' \gg E$ anyway, so the heating rate does not depend on the rest energy E.

To fix the correct value of E_{crit}, recall that we wish to compute the rate at which the grain is entirely emptied. The grain must thus have the time to nearly fully desorb its population before cooling down. We find the critical temperature above which this happens with the help of timescales for cooling and desorption. The desorption time is $t_{des}(E) = 1/k_{des}(E)$ as stated above, and we define a radiative cooling time using a continuous approach as $t^{rad}_{cool}(E) = E/P_{em}(E)$. The energy E_{crit} is the one such that $t^{rad}_{cool} = t_{des}$ and corresponds to a temperature of 35K.

We call this approximation the Heating Cycle approximation. As shown in Figure 10 (red line), it accurately reproduces the mean population in the small grain regime. It only fails for one grain size for CO, in the transition between the small and intermediate grain sizes regime. The Heating Cycle approximation does not provide an estimate for bigger grain sizes because fluctuations are not important enough to reach E_{crit}, and $t_{heat} \to \infty$.

Intermediate regime and Mean k_{des} approximation. The intermediate regime (illustrated by a $21\mathrm{nm}$ grain on Figure 10) corresponds to grains which are affected by fluctuations but never completely emptied. Their conditional mean population $\langle N|T \rangle$ is surprisingly constant. The explanation is as follows: as stated above, these grains never reach above E_{crit}, so the desorption always happens on a slower timescale than the temperature variations. The desorption mechanism thus only perceives an average desorption rate $\langle k_{des} \rangle = \int k_{des} f(E) dE$. This average process makes the population $\langle N|T \rangle$ independent of temperature, since it never has enough time to react to the fluctuations. This allows us to build another approximation for the mean population, by replacing the desorption rate by this averaged version: the population is then

$$N = \frac{k_{ads}}{\langle k_{des} \rangle}.$$

The mean k_{des} approximation is plotted in Figure 9 (green line). Once again it replicates the full computation.

Big grain regime. As the amplitude of temperature variations decreases and the energy PDF sharpens, the mean desorption rate $\langle k_{des} \rangle$ gets closer to the value at equilibrium temperature $k_{des}(T_d^{eq})$. Moreover, as can be seen by the $300\mathrm{nm}$ grain on Figure 11, the possible grain temperatures are in the photodesorption-dominated regime, so the width of the PDF has no impact since the desorption rate is constant.

Summary. The stochastic model highlights that the temperature fluctuations of small grains trigger desorption spikes that regularly completely empty grain surface. This affects very small grains ($< 2\mathrm{nm}$) for H_2O but a bigger range ($< 20\mathrm{nm}$) for CO. This leads to the rate equation at thermal equilibrium treatment to severely overestimate the size of ice mantles of these grains (in monolayers)

by up to 6 orders of magnitudes. It is possible to build approximations for the small emptied grains and the transition to the unaffected big grains, to avoid the full stochastic computation.

2.4.3. *Impact of the radiation field on the population*

Now that all the stochastic effects on the population have been understood, we discuss the results of the $\chi = 100$ and $\chi = 10^4$ models from before. The H_2O and CO populations in function of grain size are plotted on Figure 11, for the equilibrium rate equation solution N^{eq} and the true mean population $\langle N \rangle$ (as always, in monolayers). The major effect of the extinction is to reduce the flux of UV photons, cutting down on the efficiency of photodesorption and allowing larger mantles to form. As stated above, the size of ice mantles at high A_V is unrealistic because we did not take into account the freeze-out of the gas (since we fixed the gas density). The real value can only be obtained by a full computation of the gas-grain equilibrium, but this would only scale the lines in Figure 11 by a constant factor.

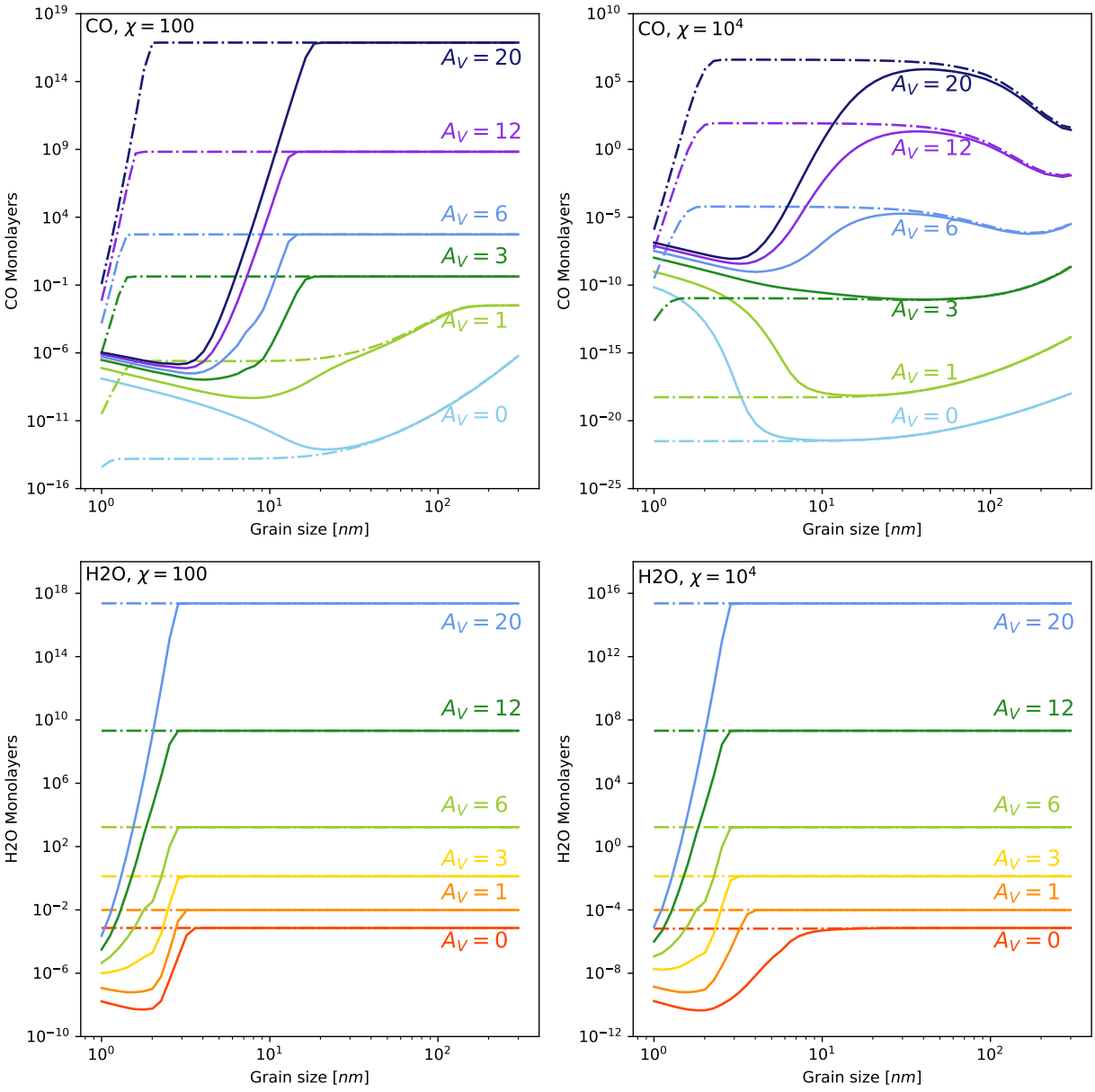


Figure 11: Mean population (plain) and equilibrium population (dotted) on grain in function of grain size (in monolayers), for different extinctions A_V and $\chi = 100$ (left), $\chi = 10^4$ (right). Top is CO ice, bottom H_2O ice.

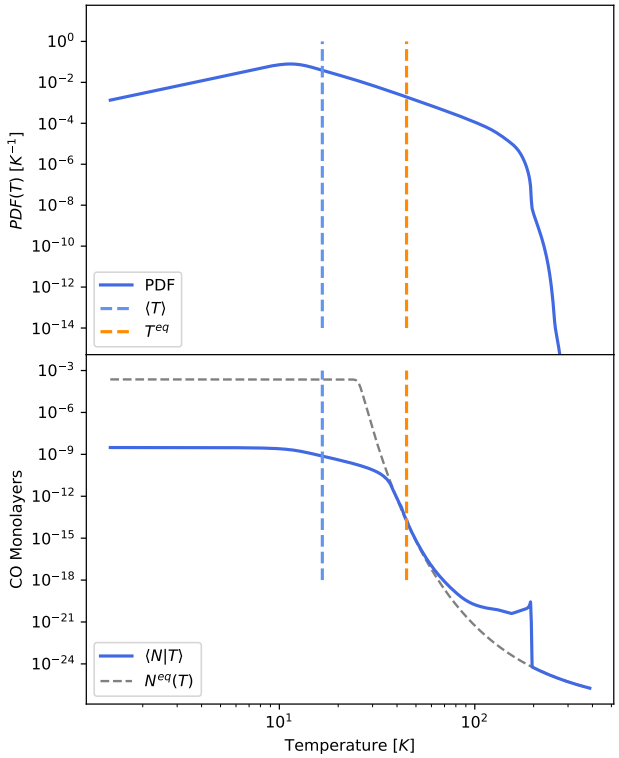


Figure 12: Temperature PDF of a $a = 2\text{nm}$ grain, under standard ISRF, $\chi = 100$, $A_V = 0$ (top) and conditional mean population $\langle N|T\rangle$ of the same grain (bottom). The mean and equilibrium temperatures are displayed.

As we have seen before that stochastic effects on the temperature are always relevant even deep in clouds, the ice mantles are also affected on the full range of extinctions. Smaller grains cannot form ice mantles because they are too frequently heat up and emptied. As in the standard ISRF case, the range of sizes affected is bigger for CO than H_2O, because of its lower binding temperature. Only very small grains ($< 2\text{nm}$) are impacted for H_2O ices. For CO ices however, the range of sizes impacted ($< 10\text{nm}$) makes it mandatory to take into account these effects rather than relying on the rate equation equilibrium treatment.

We now cover smaller effects that are still worth discussing. At $A_V = 0$, the effect of χ are reversed between CO and H_2O. For H_2O, higher χ means hotter grains, so the fluctuations can surpass $T^{switch} = 92\text{K}$ and trigger photodesorption. We thus see that a wider range of grains is affected by fluctuations for $\chi = 10^4$ than for $\chi = 100$. The effect is the opposite on CO, because in both cases the temperature are above T^{switch} anyway. The stochastic effects on population thus begin as soon as the stochastic effects on temperature do, and the latter happens at a smaller size of grains for $\chi = 10^4$ than for $\chi = 100$ (as discussed in the previous section).

The only setting that does not showcase the same three regimes as the standard ISRF case is the low A_V models of CO ice. Indeed, the $A_V = 0$ curve for $\chi = 100$ and the $A_V = 0,1,3$ curves for $\chi = 10^4$ showcase a different impact of temperature fluctuations: the mean population is actually higher than the predicted one. This can be understood thanks to Figure 12: in these settings, the grains are hot enough for their mantle to be regulated by thermal desorption at equilibrium. In the stochastic treatment, although there are temperature spikes that do boost desorption, the grains spend most of their time at temperatures lower than equilibrium. The latter effect is dominant and the grain mantle is actually bigger than the rate equations predict. However, in such conditions of strongly illuminated cloud edges, it is unlikely that any CO mantle would form anyway: the carbon in the gas-phase is mostly under atomic form C rather than molecular CO at very low A_V.

Summary. Since temperature fluctuations of small grains persist even at high extinction, the surface of these grains keeps being regularly emptied at high A_V. This means that a significant part of the available surface on dust cannot be covered in ice mantles. The rate equation at thermal equilibrium approach ignores this effect and hence overestimates the size of the ice mantles on small grains and their growth as one gets deeper in the cloud.

2.4.4. Conclusions of the model

To sum up the results obtained by the stochastic treatment of ice mantles in UV-illuminated regions, we have seen that grains smaller than 20nm are subject to temperature fluctuations, as it had already been shown in the literature [31, 28]. However, we additionally discovered that these fluctuations keep being relevant deeper in clouds, even at high extinction A_V. We showed how these fluctuations

affect the formation of ice mantles on dust: a range of grain sizes is prevented to form mantles, which increases with the volatility of the species. If for H_2O only very small grains ($< 2\mathrm{nm}$) are affected, CO mantles cannot form on grains up to $10\mathrm{nm}$. This represents a considerable fraction of interstellar dust, as small grains are the most abundant. The rate equation approach assuming thermal equilibrium ignores these effects and incorrectly predicts the same mantle size for almost all grains, largely overestimating the total density of adsorbed species. To correct the rate equations, we developed approximations for the different regimes of surface population. For the most important smaller sizes, they do not even require the computation of the energy PDF of the grain.

As a transition to the next astrophysical question in this work, our results have recalled that deeper in clouds, photodesorption becomes inefficient and the gas-grain equilibrium shifts towards a complete freeze-out of the gas phase onto the grains. Among the mechanisms that prevent this complete freeze-out, cosmic rays can provide the lacking energy necessary to heat the grains and desorb species from the mantles. In the sequel, we are interested in including the deposition of energy on the grains by cosmic-rays to our model.

3. Cosmic-ray-induced grain heating

3.1. Interactions of cosmic-rays and grains

3.1.1. *Physical processes: sputtering and whole grain heating*

As we explained in the introduction, deep in clouds, the gas and grain temperatures significantly drop. Cold grain surfaces ($\sim$10K) mean that thermal evaporation is inefficient at these depths, as is photodesorption, because UV photons are rare. The gas would, without other desorption processes, entirely freeze in the ice mantles. To explain the observations of gas-phase species in molecular clouds, including ones that are exclusively formed through surface chemistry, other desorption mechanisms must be at play. Several authors have studied cosmic rays (CR) as candidates for such so-called *non-thermal* desorption processes [24, 22, 25, 18, 41].

Cosmic rays are atomic nuclei accelerated to relativistic energies, that permeate the interstellar medium and provide energy to the gas or grains through a variety of interactions.[5] Cosmic-rays hitting a grain will deposit a fraction of their energy. The energy transfer from the cosmic ray to the grain happens in several steps [22]: the energetic particle excites electrons, that in $10^{-11}\mathrm{s}$ relax their energy in vibrations of the grain's lattice, creating a local cylindrical hot spot in the cosmic ray's wake. From this hot spot material can desorb, in a mechanism referred to as *sputtering*.

The heat then diffuses to the whole grain, which reaches a uniform temperature in $10^{-9}\mathrm{s}$ after the impact. The high temperatures triggers the thermal desorption of surface species from the mantle, and the grain loses some of its population during its cool down back to low temperatures. This second desorption mechanism is the so-called *Whole Grain Heating* (WGH)[6]. It is on this process that we will concentrate in the sequel. Before extending our stochastic model to include cosmic rays, we present how WGH-induced desorption is modelled in rate equation approaches.

3.1.2. *Rate equation treatment of Whole Grain Heating*

To include the desorption induced by WGH, most astrochemical numerical models [39, 43, 44, 45] rely on the rate-equation prescription by Hasegawa and Herbst [24] (hereafter HH). They suppose that the grain is brought to 70K after impact with a CR, leading to the following desorption rate per molecule:

$$k_{des,CR} = \mathcal{F}(70\mathrm{K})k_{th}(70\mathrm{K}). \tag{14}$$

Here the factor $\mathcal{F}(70\mathrm{K})$ is referred to as the *duty cycle*: it is the fraction of time the grain spends at 70K. It is defined as the ratio of two timescales: $\mathcal{F}(70\mathrm{K}) = t_{cool}/t_{heat,70K}$. The time $t_{heat,70K}$ is the interval

[5] The most important one is the ionization of hydrogen, which can initiate series of chemical reactions, to the point that most of the chemistry in colder parts of the ISM is actually governed by the ionization rate of hydrogen by CRs.

[6] It has also been envisaged that the energy from the CR could trigger a chemical explosion of the ice [42]. Such speculative mechanisms are out of the scope of this work.

between to CR impacts raising the grain temperature at 70K. HH simply fix its value to $3.1 \cdot 10^{13}$s (a value that is derived considering Fe nuclei with energies between 20 to 70 MeV per nucleon, striking a 0.1μm grain).

The time t_{cool} is the time it takes for the grain to cool down from 70K, which must be considered with care. In this setting, the cooling of the grain is actually not caused by photon emission, but by the energy it evacuates when it desorbs a large amount of its mantle. Indeed, the desorption of a single molecule evacuates an energy $kT_{bind} \sim 10^{-20}$J from the grain. Even if it is only a fraction of its thermal energy ($E \sim 10^{-16}$J for a $a = 0.1\mu$m grain at 10K), the total amount of desorbed molecules makes it a more efficient mechanism that radiative cooling[7]. The cooling time is estimated to $t_{cool} = 10^{-5}$s by HH, resulting from the desorption of the most volatile species of the ice.

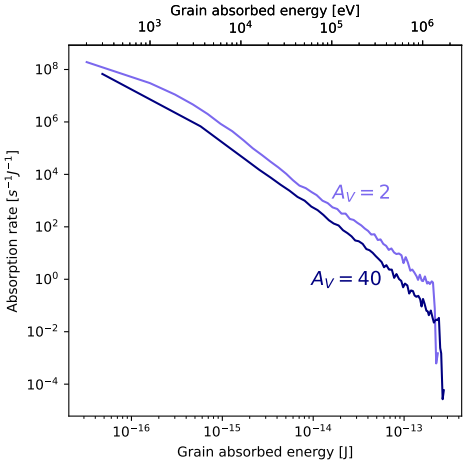


Figure 13: Absorption rate R_{abs}^{CR} for a 0.1μm grain (data from [4])

Both the duty cycle and the 70K temperature are implemented as such in most models, although fixing both values is questionable. First, the temperature at which a grain is brought is expected to widely vary depending on the energy of the CR, its constitution, and the way it hits the grain. Moreover, the interval t_{heat} also depends on the energy deposited by the CRs. Finally, the cooling time should vary with the amount of species available to desorb. This motivates us to revisit in more detail the WGH-induced desorption.

We wish to assess the accuracy of the HH formula by extending our stochastic model to include cosmic rays. WGH-induced desorption is, at its core, a desorption peak that is entailed by a temperature spike, so our model is particularly well suited to treat it. To do so, we need to add to our model the heating of the grain by cosmic rays, as well as the evaporative cooling described above.

3.2. Inclusion of cosmic-rays in the model

3.2.1. Modelling Whole Grain Heating

We now turn to the inclusion of WGH and the induced desorption in our model. Since our formalism is able to treat the desorption of the population caused by temperature fluctuations, we only have to add the heating caused by cosmic rays: its effects on the ice mantle will come as a natural consequence.

As we did for the photon absorption rate, we need to know the rate at which a CR grants a certain energy U to the grain. Computing the energy that a the CR deposits into the grain as it passes through it, and the heat diffusion that follows, is a problem of its own [46, 22]. Additionally, although less affected than the ISRF, the CR spectrum gets weaker deeper in clouds [47].

The two problems have been conjointly studied in detail by Kalvans et al. in a series of papers [2, 3, 4], in which they compute the frequency of heating events. For every considered extinction A_V, they obtain the full frequency spectrum $R_{abs}^{CR}(U)$ (in $\mathrm{s}^{-1}J^{-1}$) of deposition of an energy U in the grain. In other words, $R_{abs}^{CR}(U)dU$ is the frequency at which the grain absorbs an energy between U and $U + dU$ because of a CR impact (accounting for many possible CR constituents, and possible paths through the grain, that would yield an energy in this interval). The absorption rates they obtain is plotted in Figure 13.

We directly implemented the absorption rate in our model by extending the previous rate due to photons with the CR one at higher energies. This means that the total rate now becomes:

$$R_{abs}(U) = 4\pi^2 a^2 Q_{abs} \frac{I_U(U)}{U} + R_{abs}^{CR}(U).$$

[7]We did not consider this mechanism in section 2 because the grain had a way thinner mantle in the UV-illuminated regions we studied than in the cloud cores considered here.

Another change was that we matched the grain composition to a mixture of 60% silicates and 40% carbon to match the grain material from [4]. This makes for a slightly higher heat capacity than carbon alone.

3.2.2. *Modelling evaporative cooling*

The other mechanism we need to implement in the model is the cooling of the grain caused by the evaporation of its mantle. The sudden temperature spike upon a CR impact is expected to trigger a massive desorption of species, that will evacuate a significant amount of energy and help in the grain's cooling.

We need to revisit our prescription for the stochastic transitions caused by thermal desorption: they only affected the population in the previous section, and will now cool the grain as well. We use the shorthand notation $\Delta E = kT_{bind}$. This is the energy necessary for making one molecule exit the potential of the binding site, which means that, for every thermal desorption occurring on the surface, the grain loses an energy ΔE. Note that photodesorption, on the other hand, does not cool down the grain, since it is the photon's energy that is used instead.

Recall that X is the coupled stochastic process for the thermal energy and surface population $X = (E,N)$. The transition rate $p_{X\to Y}$ should be corrected compared to 2.2.1 so that thermal desorption transitions decrease the energy E of the grain by ΔE. We rewrite all the possible transitions for the sake of clarity:

- Transitions from E to any $E' > E$, at fixed N, corresponding to the absorption of a photon of energy $E'-E$, or the deposition of an energy $E'-E$ by a CR. This happens at a rate $R_{abs}(E'-E)$.
- Transitions from E to any $E' < E$, at fixed N, corresponding to the emission of a photon of energy $E - E'$. This happens at a rate $R_{em}(E - E')$.
- Transitions from N to $N' = N + 1$, at fixed E, corresponding to adsorption. This happens at a rate k_{ads}.
- Transitions from N to $N' = N - 1$, at fixed E, corresponding to photodesorption (thermal or photodesorption). This happens at a rate $k_{ph}(E)$.
- New transitions from (N,E) to $(N-1, E-\Delta E)$ correspond to thermal desorption. They happen at a rate $k_{th}(E)$.

The transition rate thus writes:

$$p_{X\to Y} = p^*_{E\to E'}\delta_{N=N'} + k_{ads}\delta_{N'=N+1}\delta_{E=E'} + Nk_{ph}\delta_{N'=N-1}\delta_{E=E'} + Nk_{th}(E)\delta_{N'=N-1}\delta_{E'=E-\Delta E}$$

where we added the $*$ on the rate that was denoted $p_{E\to E'}$ previously, to stress that it does not contain all the transitions affecting energy anymore. It only accounts for the absorption of energy from photons and CR and the emission of photons (the first two bullet points). Its value is:

$$p^*_{E\to E'} = \begin{cases} R_{abs}(E' - E) & \text{if } E' > E \\ R_{em}(E - E', E) & \text{if } E' < E. \end{cases}$$

The evaporative cooling makes the temperature of the grain dependent on the population. Before, it was only the population N that was dependent on E through the thermal desorption rate $k_{th}(E)$; whereas the energy PDF could be independently solved for. We will not be able to rely on the same simplification here, which complicates the mathematical and numerical resolution of the problem. We will now plug the transition rate formula in the Master Equation (8) to derive the equations for the temperature PDF and the conditional mean population.

3.2.3. *Coupled equations for the temperature PDF and the conditional mean population*

Thermal energy PDF: Previously we were able to derive a closed equation for the thermal energy PDF because the temperature fluctuations did not depend on the population. Now that we consider that desorption cools the grain, this simplification is not possible anymore. Writing the Master Equation (8) with the new rate and summing it over the population, the equation obtained for the population PDF is (see Appendix F.2 for the derivation):

$$\int_{\mathbb{R}^+} f(E')p^*_{E'\to E}dE' + f(E+\Delta E)k_{th}(E+\Delta E)\langle N|E+\Delta E\rangle = \\ f(E)\left(\int_{\mathbb{R}^+} p^*_{E\to E'}dE' + k_{th}(E)\langle N|E\rangle\right) \tag{15}$$

There is a new gain term (on the first line) corresponding to arrival at state E because of the grain cooling down from $E+\Delta E$ by thermal desorption, and a new loss term (on the second line) corresponding to the grain leaving state E by thermal desorption. The rate at which this transitions occur involves the conditional mean population.

Conditional mean population: The equation for the conditional mean population $\langle N|E\rangle$ is also affected by the inclusion of evaporative cooling in the model. It writes (see Appendix F.3 for the computation):

$$\int_{\mathbb{R}^+} \langle N|E'\rangle f(E')(p^*_{E'\to E} + f(E)k_{des}(E'))dE' + \langle N(N-1)|E+\Delta E\rangle f(E+\Delta E)k_{th}(E+\Delta E) = \\ \langle N|E\rangle f(E)\left(\int_{\mathbb{R}^+} p^*_{E\to E'}dE' + k_{des}(E)\right) + \langle N(N-1)|E\rangle f(E)k_{th}(E) \tag{16}$$

The system formed by (15) and (16) is not closed. Indeed, the equation for the population depends on the second moment of N through the term $\langle N(N-1)|E\rangle$. It expresses the flux of population when a thermal desorption occurs: if we are at state $E+\Delta E$, with a population N, then upon thermal desorption (which occurs at a rate $Nk_{des}(E+\Delta E)$), the grain is brought to E with a population $N-1$. The flux of population is thus given by $(N-1)Nk_{des}(E+\Delta E)$.

Approximation of the second moment: To try and close the system, we could derive an equation for $\langle N^2|E\rangle$, but it would involve the third moment $\langle N^3|E\rangle$, and so on. This is a common problem when deriving equations on moments of distributions. The way to close the system to a finite number of equations is to approximate one moment by a lesser order. We will keep only two equations and approximate $\langle N(N-1)|E\rangle$ as a function of $\langle N|E\rangle$. We suppose that if $\langle N|E\rangle \leq 1$, the most likely values of N are $N=0$ or $N=1$. In either case the product $N(N-1)=0$, so we approximate $\langle N(N-1)|E\rangle \simeq 0$. Otherwise, if $\langle N|E\rangle \geq 1$, we neglect the variance of N, so that $\langle N^2|E\rangle \simeq \langle N|E\rangle^2$, meaning that $\langle N(N-1)|E\rangle \simeq \langle N|E\rangle(\langle N|E\rangle - 1)$. The final approximation that we make is thus

$$\langle N(N-1)|E\rangle \simeq \langle N|E\rangle \max(\langle N|E\rangle - 1, 0) \tag{17}$$

The correction at $N<1$ also has the advantage of preserving the positivity that $\langle N(N-1)|E\rangle$ exhibits. The validity of this approximation is evaluated in Appendix D.

Plugging the approximation in equation (16) closes the system { (15), (16) }. However, it is highly non-linear, which makes its resolution still more complicated than solving an eigenproblem. It had not been encountered before in stochastic models of grain surface processes, so we had to develop new resolution methods. We relied on an iterative numerical method which we implemented in `StoCapy` and that proved to offer satisfying results. It is presented in the Appendix (G.3) to avoid plaguing this presentation with technical details.

Finally, we stress that the conditional mean population is a priori not simply proportional to the gas conditions anymore (although the non-degenerate parameter is still $n_{gas}(\mathrm{A})\sqrt{T_{gas}}$). This is expected as the temperature PDF is now dependent on gas conditions, because of the evaporative cooling. This means we will need to explore a range of gas temperatures and densities.

3.3. Results

3.3.1. Model parameters

We now showcase the results of the model. To center the presentation on the most interesting points, we will keep the parameters of the model fixed when they did not showcase a behaviour deemed worth discussing.

Grain size. The grain parameters were mostly limited by the data available for the CR absorption rates provided in [4]. Data is given for three grain sizes are $a = 50\text{nm}$, 100nm, and 200nm. Since the results are qualitatively similar, we will discuss the $a = 100\text{nm}$ size (the common value used in single-size astrochemistry models). All these grains are large enough to be at thermal equilibrium in the absence of cosmic rays.

Mantle species. We consider CO as a species: more volatile than H_2O, it should be the one desorbing most from realistic mixed-ices. This also allows for comparison with Léger et. al. [22] where it is considered to be the coolant species.

Radiation field. In molecular cloud conditions, the UV photons are heavily extinguished so the scaling factor χ on the UV field has little impact. We keep it fixed to $\chi = 1$. Extinctions from $A_V = 2$ to $A_V = 40$ were considered, which affects the equilibrium temperature of the grain, the photodesorption rate, as well as the CR absorption rates. However, the effects were only quantitative and as expected : the mantles get thicker at higher A_V. For the purpose of keeping our discussion short and to the point, we will fix a typical extinction $A_V = 9$ in the sequel.

Gas conditions. Since the mean population is no longer simply proportional to $n_{gas}(\text{CO})\sqrt{T_{gas}}$ (although it still is the non-degenerate parameter), we have to explore the gas conditions. In fact, they proved to be the most important parameter, and dictate the boundaries between several regimes with widely different results. We fix the gas temperature to $T_{gas} = 10K$ (a typical cloud core condition) and only vary the density. Typical proton densities in a molecular cloud are $n_H = 10^4 - 10^6\text{cm}^{-3}$, and the abundance of CO (ratio of CO density to n_H) can vary from 10^{-4} (if all carbon is in the form of CO gas) to very low values such as 10^{-19} (because of the freeze-out of CO onto grains). We considered gas densities of CO from $n_{gas}(\text{CO}) = 0.5\text{cm}^{-3}$ to $n_{gas}(\text{CO}) = 300\text{cm}^{-3}$. The lower bound might seem not low enough, however we will see that the population becomes proportional to $n_{gas}(\text{CO})\sqrt{T_{gas}}$ again at low density so the results can in fact be extrapolated. The higher bound of 300cm^{-3} might be a bit high as well, but was chosen so that the high-density limit regime of the model was fully covered.

3.3.2. General behaviour and comparison to literature rates

We present the general behaviour of ice mantles under the influence of cosmic rays, depending on the gas density (which dictates the adsorption rate k_{ads}). The left panel of Fig. 14 displays the full computation of the mean population $\langle N \rangle$ (in blue dots, expressed in monolayers) in function of gas density. We also showcase the results without evaporative cooling (green plain line) (amounting to letting $\Delta E = 0$ in the equations) to see its impact, and the ice mantle the grain would have without cosmic rays (orange plain line). Two limit regimes and a transition regime can be highlighted. At low density (for $n_{gas}(\text{CO}) \leq 2\text{cm}^{-3}$) the mean population is the same with or without evaporative cooling. On the contrary, at high density ($n_{gas}(\text{CO}) \geq 50\text{cm}^{-3}$), the mean population is the same as the one without cosmic rays at all. At intermediate densities, the mean population transitions with a sharp slope between the two asymptotic behaviours, which are separated by several orders of magnitude. The right panel of Fig. 14 showcases the desorption rate per molecule on the grain for the same models. It is given by $k_{ads}/\langle N \rangle$.[8]

We compare our results to literature rates on Figure 14, beginning by the usual and widely used prescription by HH that we already introduced.

Hasegawa and Herbst 93 . [24] The prescription by HH (Eq. (14)) is plotted in Fig. 14 (green dotted line). We see that it does not approximate correctly any of the two regimes, and in fact does not even

[8]Since at statistical equilibrium the mean number of desorptions per second is equal to the number of adsorptions per second k_{ads}, the desorption rate *per molecule* is indeed given by $k_{ads}/\langle N \rangle$.

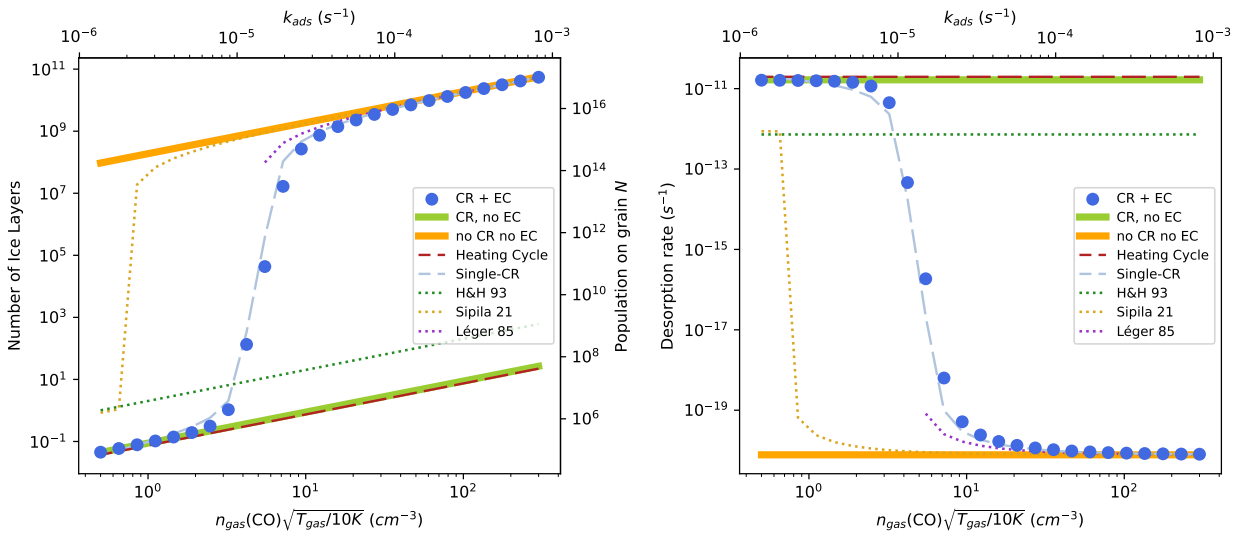


Figure 14: Population (in layers) (*left*) and total desorption rate per molecule (*right*) in function of the gas density and adsorption rate, on a $a = 100\text{nm}$ grain (under $A_V = 9$, $\chi = 1$). CR means Cosmic Rays were included, EC means Evaporative Cooling was included (see the text for details). For the approximations (dashed and dotted), see text.

account for the existence of two regimes. It overestimates the population by an order of magnitude at low CO density, and underestimates it by a factor 10^8 at high CO density. Although the numerical values used by HH are founded on a different CR spectrum, we see that even if one re-evaluated the value of the duty cycle or the 70K temperature, the formula could at best approximate one regime or the other. The inaccuracy of this formula has far-reaching implications, as it is implemented as such in many numerical models of astrochemistry.

Sipila et al. 2021. [25] (hereafter Sipila), uses the same base prescription as HH but revisits the value of the duty cycle $\mathcal{F}(70\text{K})$. They use different time intervals between heating events to 70K than HH, based on more recent CR spectra. They compute a cooling rate that is not fixed but instead based on the population available for evaporative cooling on the grain. They also add a correction to the rate so it does not allow to desorb twice from the same site in the same heating event, should the cooling time be too long. Their prescription writes after adaptation to our case:

$$k_{des,CR} = \frac{1}{t_{heat}} \min\left(\frac{E(70\text{K})}{\Delta EN}, 1\right)$$

This is plotted in Fig. 14 (golden dotted line). It accounts for the existence of two regimes but the position of the transition is not correctly identified, and its slope is too steep. It overestimates the population by several orders of magnitude around $n_{gas}(\text{CO}) \simeq 1\text{cm}^{-3}$ because it misses the change of regime. The low density limit is overestimated by a factor of 10, as HH.

Léger et al. 85. [22] (hereafter Léger)[9] Their estimate of the desorption rate consists in counting the number of desorptions necessary to cool down from a peak temperature E_{CR} to 25K, temperature under which the radiative cooling is assumed to dominate. This number is then multiplied by the rate $R_{abs}^{CR}(E_{CR})$ at which the grain is brought at this energy. It writes as a desorption rate per molecule:

$$k_{des,CR} = \frac{1}{N} \int_{E(25K)}^{\infty} R_{abs}^{CR}(E_{CR}) \frac{E_{CR} - E(25K)}{\Delta E}.$$

It accounts for a whole range of temperatures after CR impact, rather than the fixed 70K. It is showcased on Fig. 14 (violet dotted line). It better locates the transition than Sipila. It overestimates slightly the population because it does not account for the desorption that can occur during the radiative cooling of the grain, from $\sim 25K$ to equilibrium temperature. Most importantly, when k_{ads}

[9] Léger initiated the study of cosmic-ray induced desorption, with an extensive study of the CR-grain interaction. The timescales HH and several other authors rely on are actually based on the detailed physics covered by Léger.

becomes less than $Nk_{des,CR}$, the approximation is not defined. This corresponds to the grain not having enough population to cool down to 25K purely through evaporation.

As a summary, none of these rates faithfully reproduce the behaviour of the mean population in function of gas density. The widespread HH prescription does not acknowledge the presence of several regimes, leading to errors of several orders of magnitudes. To better understand the physical reasons behind their inaccuracies, and in view of building satisfying approximations, we now conduct a detailed analysis of the different regimes.

3.3.3. *Detailed study of the three regimes*

Low density regime. We begin by analysing the behaviour of the grain for $n_{gas}(\mathrm{CO})$ under $2\mathrm{cm}^{-3}$, the regime in which the evaporative cooling has no effect on the size of ice mantles. The top panel of Figure 15 displays the temperature PDF of a $a = 100\mathrm{nm}$ grain for $n_{gas}(\mathrm{CO}) = 0.5\mathrm{cm}^{-3}$ (blue curve).

The majority of the PDF is located in the sharp bump around its equilibrium temperature at rest of 13K (that is to say, the equilibrium temperature computed with just photons and no CRs). Without cosmic rays, a $a = 100\mathrm{nm}$ is big enough to be well represented by an equilibrium temperature, and the bump here is the distribution of temperatures when the grain is at rest, not having been hit by a CR in a long time. The PDF exhibits a long tail, which extends to over a hundred Kelvin and then abruptly cuts. It represents the grains having recently received energy from a CR and cooling down to lower temperature. The cut-off corresponds to the maximal energy in the R^{CR}_{abs} spectrum, around $2 \cdot 10^{-13} J$: It is the temperature a grain is brought to in the CR impact scenarios that provide the most energy. Those are caused by the heavier CRs (such as Fe nuclei) with the highest energy, passing right through the center of the grain. Beyond the cut off, the PDF is at very low values because the scenarios bringing the grain to these temperatures involve being hit twice in a row by a CR.

Now, concerning the population on the surface, the middle panel of Fig. 15 (blue curve again), which presents the conditional mean population, shows that it is heavily affected by the rare but very high temperature spikes. In-

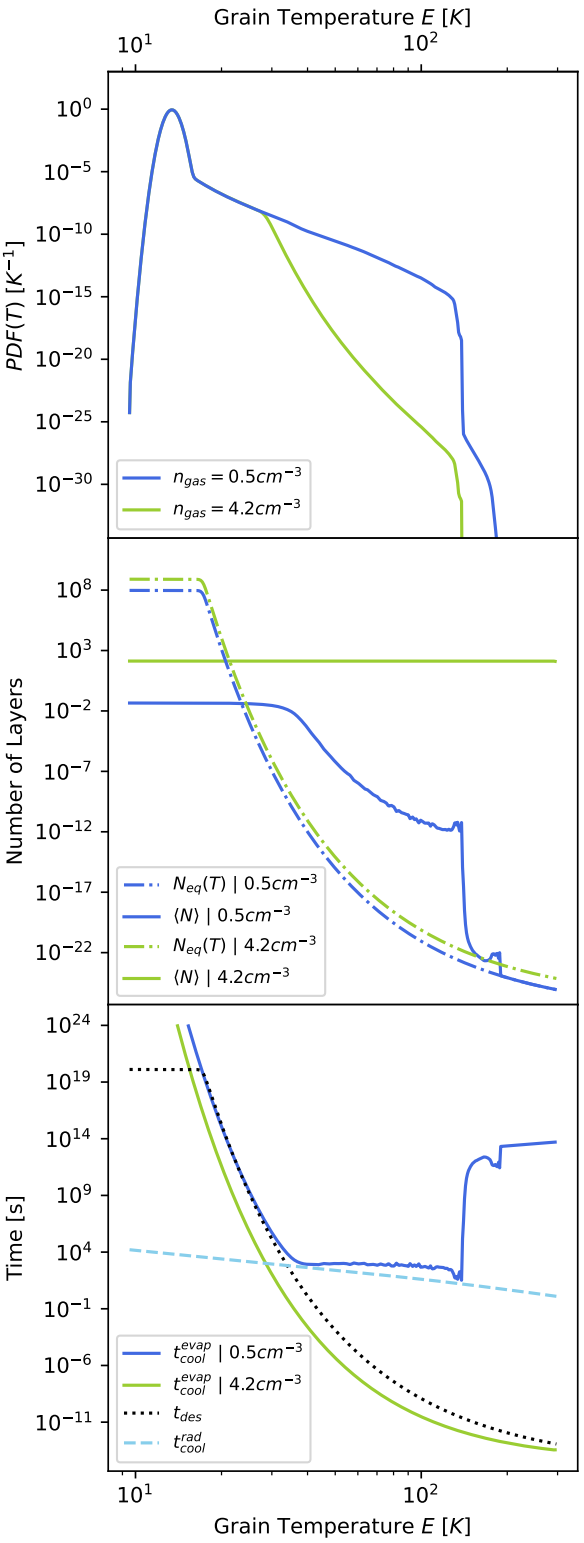


Figure 15: Temperature PDF (*top*), conditional mean population $\langle N|T\rangle$ (plain) and equilibrium population $N^{eq}(T)$ (dotted) (*middle*), evaporative timescale (*bottom*), for two CO gas densities (blue: $n_{gas} = 0.5\mathrm{cm}^{-3}$ and green: $n_{gas} = 4.2\mathrm{cm}^{-3}$). The radiative and desorption timescales on the bottom panel are independent of gas density.

deed, the low population plateau is far from the equilibrium value predicted by rate equations (N^{eq} dotted line). The picture for the grain hit by a CR is similar to the case of small grains hit by a UV photon (cf the discussion on the blue curve of Fig. 10): When the grain is brought to high temperatures, it loses a significant part of its surface population while cooling down. Grains hit by two CR in a row are almost completely emptied because desorption is extremely fast at theses temperatures. Since the grain comes back to low temperatures with very little molecules, we can in fact consider that it is completely emptied by sufficiently energetic CR impacts.

The final point to discuss is why the evaporative cooling has no impact in this low density regime. To explain this, the bottom panel of Fig. 15 showcases the timescales of the system. The radiative cooling time is, as earlier, computed as $t_{cool}^{rad} = E/P_{em}(E)$ (where P_{em} is the radiative power, emitted in photons). The desorption timescale is the timescale for the surface population evolutions, $t_{des} = 1/k_{des}$. It is not the same as the evaporative cooling timescale, because the latter depends on the amount of energy evacuated at each desorption. We define it as the ratio between the energy E and the evaporative power $P_{em}^{evap} = \Delta E \langle N|E \rangle k_{des}$. This is the mean energy evacuated by the whole grain per second. The evaporative cooling timescale is thus $t_{cool}^{evap} = E/\Delta E \langle N|E \rangle k_{th}$. These three timescales are plotted on Fig. 15. We can see that the radiative cooling is always faster than the evaporative cooling (blue curve, even though by a small margin). The population on the grain is simply not large enough for the power lost through desorption to be significant compared to the radiated one. In addition, almost everywhere, t_{cool}^{evap} is also longer than t_{des}, which means that the grain reaching population equilibrium would not be enough to cool the grain. Indeed, when photodesorption is negligible, $t_{cool}^{evap} > t_{des}$ is equivalent to $E/\Delta E > \langle N|E \rangle$: this simply expresses that the number of molecules needed to evacuate the energy E is larger to what is in average available on the grain at this energy.

Heating Cycle approximation for the low density regime. Since the grains are emptied by the CR impacts, one can rely on the Heating Cycle approximation to estimate the mean population $\langle N \rangle$ on the grain. The absorption rate in equation (13) from Section 2 simply now includes the energies deposited by CR. The temperature T_{crit} above which the grain empties because desorption becomes faster than radiative cooling is still 35K as in Section 2. It is the intersection of t_{cool}^{rad} and t_{des} in the bottom panel of Fig. 15. The results of the Heating Cycle approximation are on Fig. 14 (red dashed line): we see that it indeed perfectly reproduces the low density regime.

Intermediate regime. As the gas density increases, so does the adsorption rate and the mean population on the grain. In the time interval between CR heating events that empty the surface, it can accumulate a larger population. Hence, the more species desorb upon CR impact. The instantaneous cooling power due to desorption is increased, until it becomes more efficient than the emission of photons and begins to play a role. This happens around $n_{gas}(\mathrm{CO}) = 2.5\mathrm{cm}^{-3}$ on Fig. 14.

The intermediate regime is illustrated with the results for $n_{gas}(\mathrm{CO}) = 4.5\mathrm{cm}^{-3}$ (green curve) on Figure 15. We can see that the conditional mean population is almost independent of temperature. This does not mean that CR do not have an impact, because the plateau value is still lower by several orders of magnitude than the equilibrium one at around 10K. The explanation is the following: the number of desorptions from the mantle upon WGH is large enough to quickly evacuate a significant amount of energy, before the population itself significantly changed. This is attested by the timescales on the bottom panel: the evaporative cooling time is shorter than the desorption time. The grain cannot be completely emptied anymore, as it has necessarily returned to low temperatures after only losing a part of its mantle.

The evaporative cooling being faster than the radiative cooling translates to the PDF: it displays a steeper slope in its tail compared to the previous regime (blue curve). Since the grain cools down faster, it is indeed less likely to be observed at these temperatures, hence the lower values.

The radiative cooling becomes efficient again only under 27K, and this is not because of a lack of species to desorb, but because desorption is simply ineffective at these low temperatures. To summarize this regime, the picture is the following: upon a temperature spike, the grain desorbs the number of molecules required to evacuate the surplus energy and cool down to 27K, from which point the

grain cools down radiatively to its equilibrium temperature (it can still desorb molecules during this phase, but not enough to provide a significant cooling compared to photon emission).

High density regime. When the grain population at rest is far greater than the number of desorbed molecules required to cool down, the grain becomes basically unaffected by the temperature spikes. It will only lose a negligible amount of its mantle, that is enough to cool the grain down to equilibrium. This explains why the intermediate regime eventually merges with the equilibrium population curve without cosmic rays around $n_{gas}(\mathrm{CO}) = 40\mathrm{cm}^{-3}$ (see Fig. 14).

Transition point. There is actually a very narrow transition between the low density and intermediate regime, which corresponds to the $n_{gas}(\mathrm{CO}) = 2.5\mathrm{cm}^{-3}$ point in our grid of models. It displays an interesting intermediate situation between the blue and green curves of Fig. 15, as the evaporative cooling is only efficient for a very short time before being supplanted by radiative cooling. We avoided covering it here to shorten the presentation, but it is discussed in Appendix B.

Summary. For gas densities lower than $n_{gas}(\mathrm{CO}) = 2.5\mathrm{cm}^{-3}$, we find that the adsorption rate is too slow to build ice mantles. The cosmic ray impacts completely empty the grain, and its population is well estimated by the Heating Cycle approximation. On the opposite, for densities higher than $n_{gas}(\mathrm{CO}) = 2.5\mathrm{cm}^{-3}$, the grain accumulates a larger population between CR impacts. At each heating event, the grain loses the number of molecules required to make it cool down to lower temperatures, at which radiative cooling dominates until the grain returns to its equilibrium temperature. Since the mantles grow with the adsorption rate, but the number of molecules required to cool down is constant, the impact of cosmic rays becomes increasingly negligible as the gas density increases. We now make use of this physical insight to build an accurate approximation of the intermediate to high gas density regime, to complement the Heating Cycle approximation.

3.3.4. Single-CR approximation

The approximation we build is founded on the timescales of the system. Indeed, the full master equation allows all possible transitions for all states, which makes it impossible to solve analytically. In practice, however, some transitions are very unlikely and can be neglected. For instance, the timescale for adsorption is (for the regimes where CRs have an impact on population) $1/k_{ads} \geq 10^4$s (see the top scale on Fig. 14), while the cooling time stays mostly under 10^4 second. This means there should be around one adsorption only during the cool down of a grain after a heating event, which can be neglected. Identifying the important transitions, one can derive a simplified master equation for the tail of the PDF. It will then be used to find an approximation for $\langle N \rangle$. The full computations are made in Appendix C, here we present the physical insight.

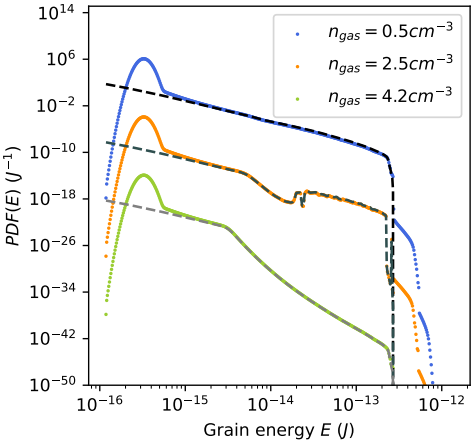


Figure 16: Approximation of the PDF tail for three gas conditions. The lines were vertically shifted for better readability.

At the temperatures over which the PDF tail extends, we can make the assumption that the grain only cools down (through radiative and evaporative cooling) and has no time to adsorb new molecules. We also neglect the unlikely probability that it is heated again by a second CR, hence calling this approach the Single-CR approximation. The probability $f(E)dE$ of finding a grain at an energy in $[E, E + dE]$ is the time dt spent in this energy interval times the frequency at which it reaches this interval. The time dt spent in the interval is the time necessary to cool down from $E + dE$ to E, $dt = dE/(P_{em}^{rad}(E) + P_{em}^{evap}(E))$. It will cross this interval for every heating event that brings him to or above this energy, so at a frequency $\int_E^\infty R_{abs}(E')dE'$ (once again we neglect the equilibrium energy the grain has at rest). This

yields the approximation for the PDF tail, plugging the expression of the evaporative power:

$$f_{tail}(E) = \frac{\int_E^\infty R_{abs}(E')dE'}{P_{em}^{rad}(E) + \Delta E \langle N|E\rangle k_{th}(E)} \tag{18}$$

To show the accuracy of this expression, we plot the approximation and the full PDF for three regimes (radiative-dominated, transition regime, and evaporative-dominated) on Fig. 16. Remark that it stops at the first cut-off of the PDF since we ignore multiple-CR events. Although accurate, this approximation as it is relies on the conditional mean $\langle N|E\rangle$ (and its values have been used to plot the dotted lines in Fig. 16), so it is not of much use as such.

What we want to obtain now is a closed formula for $\langle N|E\rangle$, and for this we will make further assumptions. Since we already have the Heating Cycle approximation for the low-density regime, we are interested in the intermediate transition and the high density regime. In those, the evaporative timescale t_{cool}^{evap} is faster than the timescale for population changes t_{des}, so the surface population has little time to adapt to temperature changes, and the conditional population $\langle N|E\rangle$ is almost constant. We thus make the simplifying assumption $\langle N|E\rangle \simeq \langle N\rangle$. The population being independent of the energy, it allows us to write the formula[10]:

$$\langle N\rangle \int f(E) k_{des}(E) dE = k_{ads}, \tag{19}$$

simply expressing that the mean total desorption must be equal to the adsorption at statistical equilibrium. We only need an expression for the full PDF $f(E)$.

We approximate the PDF using (18) for the tail and modelling the bump at low energies by a Dirac function at the equilibrium energy:

$$f(E) = \delta_{E=E^{eq}} + f_{tail}(E) \tag{20}$$

This means that, given a $\langle N\rangle$ value, one has an approximation for the PDF using Eq.(20), which can be used to derive a new value $\widetilde{\langle N\rangle}$ using this time Eq. (19). We end up with a fixed point equation: it should hold that $\widetilde{\langle N\rangle} = \langle N\rangle$.

Numerically, we relied on dichotomy to find the solution for the sake of robustness, but other methods could be implemented. From the estimate of the mean population $\langle N\rangle$, one can have the equivalent desorption rate per site $k_{ads}/\langle N\rangle$.

The single CR approximation is plotted on Fig. 14. It manages to reproduce the slope and locate accurately the transition between the two regimes. This is because it encompasses the best of the previously detailed rates: it acknowledges the full spectrum of CR deposited energies rather than a single temperature, and also accounts for the desorption that can occur at the end of the grain cooling down, once radiative cooling takes over (that [22] neglected).

3.3.5. Conclusions of the model

To our knowledge, a master equation approach of Whole Grain Heating by cosmic rays had never been conducted before in the literature. New methods were developed to solve the systems of non linear equations that stemmed from the inclusion of evaporative cooling in the model. Our stochastic model allowed us to assess the usual rate prescribed by Hasegawa and Herbst [24] on which many numerical models rely. We showed that this prescription overestimates the size of ice mantle by an order of magnitude at lower gas densities, and underestimates their size by several orders at higher gas densities. It fails to describe the two regimes on which we shed light with our full computation.

We found that for gas densities lower than $n_{gas}(\mathrm{CO}) = 2.5\mathrm{cm}^{-3}$, the cosmic rays completely empty the grain, and we validated the Heating Cycle approximation in this regime. For densities higher than $n_{gas}(\mathrm{CO}) = 2.5\mathrm{cm}^{-3}$, the grain has a high enough population for evaporative cooling to dominate. The WGH-induced desorption in this regime is the sum of two contributions: the first

[10] Otherwise $\langle N|E\rangle$ is below the integral sign, which yields back equation (11).

is the necessary number of desorptions required to cool down the grain from its peak temperature to an intermediate one, at which radiative cooling dominates again. The second contribution is the desorption of species during this radiative cooling phase. We built the accurate Single CR approximation for this regime. Combined with the Heating Cycle one, it allows to fully replicate the master equation results on the entire range of gas conditions and could be used as a better alternative to what is usually implemented in models.

4. Conclusion

In this work, the ice mantles on interstellar dust were studied with a stochastic formalism, that took into account what is often neglected in models: the fluctuations of dust grain temperatures. Based on the methods of Bron et. al. [31], who only studied the formation of H_2 on grains, we extended their master equation approach to the study of ice mantles.

We first focused on the formation and destruction of ice mantles in strongly illuminated environments, such as cloud edges or protoplanetary disks. For this, we derived the thermal energy distribution of grains. We showed that the progressive extinction of UV light does not let small grains reach equilibrium, as their colder temperature makes them all the more sensitive to fluctuations. These fluctuations were found to prevent the formation not only of volatile CO mantles, but also, to a lesser extent, of more refractory H_2O mantles. The range of sizes that cannot form ice mantles is particularly significant in the case of CO ($a < 10\text{nm}$) and is expected to affect the gas-grain equilibrium. Such behaviours were not predicted by the usual equilibrium approach used in rate-equation models, and we proposed approximations to improve their accuracy.

In a second time, we studied the desorption of ice from grains that is caused by cosmic rays. Understanding the non-thermal desorption mechanisms is necessary to explain the gas-phase observations of cold molecular clouds, and the interaction of cosmic-ray with grains might be the most important one. The gas-grain chemistry of these dense regions is of prime relevance in the context of star and planet formation. At the exception of Monte-Carlo simulations, the desorption mechanism of Whole Grain Heating by cosmic rays had never been studied in a stochastic formalism. We extended the master equation approach to include the heating of grains by cosmic rays and the evaporative cooling. Our work proved Whole Grain Heating to be a complex mechanism that is not described with enough accuracy in astrochemistry models. We showed that a precise modelling must consider the full spectrum of cosmic rays and the wide range of energies they can impart to an interstellar grain. The usual prescription made in models, relying on a single peak temperature and fixed timescales, is incorrect by several orders of magnitude. At a low gas density, volatile mantles can be fully evaporated, more frequently than usually predicted. Because the efficiency of cosmic ray desorption abruptly drops if the grain is able to repopulate its surface between two CR strikes, a precise prescription of desorption rates is all the more necessary to understand the gas-grain equilibrium.

In our models, we had to solve new coupled systems of equations that had not been encountered previously in master equation approaches. Such coupling and non-linearity are bound to appear again as one extends the stochastic formalism to include more aspects of grain surface chemistry. This work is thus also a first step in the mathematical and numerical approach of increasingly complex stochastic models.

All the numerical computations in this work were made in the `StoCapy` Fortran code, that was developed from the ground up and became a quite extensive project (of around 5000 lines of code). In its current state, it can compute the temperature distributions and conditional populations of for a distribution of interstellar grain size, under different radiation field parameters ; and can also include cosmic rays for the settings for which data is available.

The next milestone for this study would be the implementation of our stochastic results in a full astrochemistry code. Because of the sheer scale of astrochemical simulations, this would rely on the approximations built over the course of our work. This would frame the important impact of the stochastic processes, studied here at the scale of a grain, in the bigger picture of the chemistry of interstellar clouds and of the life cycle of the ISM.

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A. On photodesorption yields

Here we discuss in more detail how to implement the experimental results on photodesorption. This complements Section 2.1.5 of the report. The experiments conducted on photodesorption measure a quantity known as the yield Y. Expressed in molecules per photon, it is the probability of desorption knowing that a photon has collided with the surface. One must carefully understand how yields are measured in laboratory experiments to correctly implement photodesorption in a low-population setting. Experiments consist in sending photons on a macroscopic surface covered *with several layers of ice*, and counting the number of desorbed molecules. This means that it reproduces a regime in which the grain is also coated with many ice layers. In this regime, the photo-desorption rate is simply the product of the number of UV photons hitting the surface per second multiplied by the yield. If F_{UV} is the flux of such UV photons (photons per second per unit surface), it writes

$$\left.\frac{dN}{dt}\right|_{ph} = -Y F_{UV} \pi a^2.$$

However, we are not in the regime of thick ice layers on grains. Once again, if we suppose that only the top l_{act} layers are available for photodesorption and molecules below are trapped, then the formula above holds for more than l_{act} layers on the grains, i.e. when $N > l_{act} N_s$. In the low-population regime, the naive correction to the rate is to take it to be only a $N/l_{act}N_s$ fraction of the thick ice mantle setting. This means that the corrected rate is:

$$\left.\frac{dN}{dt}\right|_{ph} = -Y F_{UV} \pi a^2 \min(1, \frac{N}{l_{act} N_s}).$$

To stay consistent with thermal desorption, we will always consider the low-population regime. Expressing the UV flux in terms of the radiation field, one finally has:

$$\left.\frac{dN}{dt}\right|_{ph} = -\frac{Y 4\pi^2 a^2}{l_{act} N_s} \left(\int_{UV} \frac{I_U(U)}{U} dU \right) N = -N k_{ph}.$$

In all generality, the yield depends on the incident photon energy U and should be under the integral signs. However, experiment data provides us with already-integrated yields for typical astrophysical conditions and it those values that are given in Table 1.

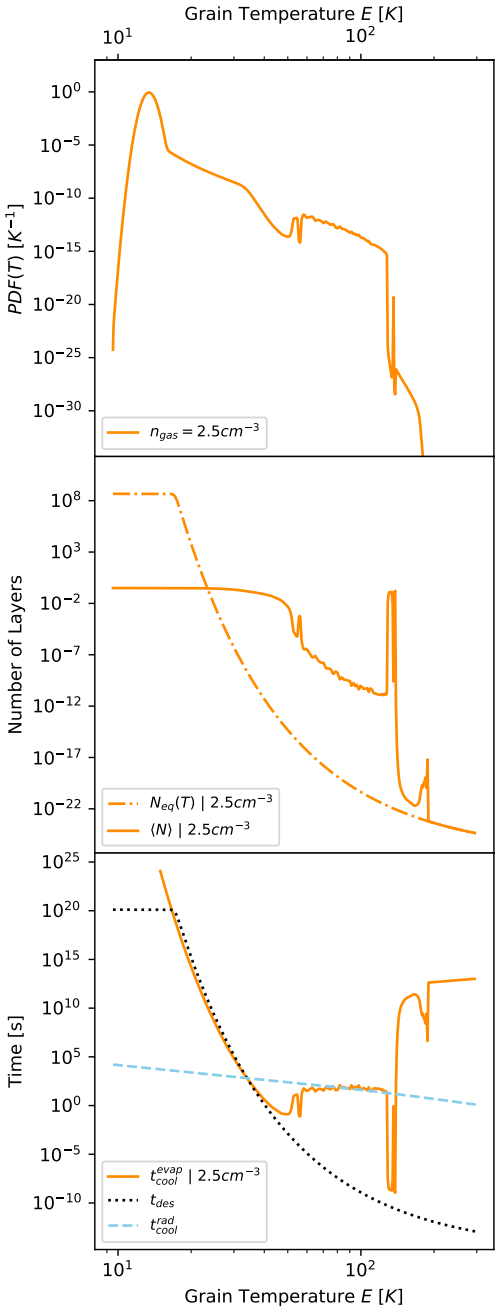


Figure 17: Temperature PDF (*top*), conditional mean population $\langle N|T \rangle$ (*middle*), cooling and desorption timescales (*bottom*), for CO gas density $n_{gas} = 2.5\text{cm}^{-3}$.

B. On the $n_{gas} = 2.5\text{cm}^{-3}$ transition point

Here we discuss the transition points between the low density and the intermediate regimes for the ice mantles under heating by CR. This complements Section 3.3.3 of the report. The orange line on the top and middle panels on Figure 17 are the PDF and conditional mean population for a gas density $n_{gas}(\text{CO}) = 2.5\text{cm}^{-3}$.

We begin by discussing the mean population. The main features are the two spikes at 150 K and 200 K, corresponding to absorption of the maximal energy in the R_{abs}^{CR} spectrum, and absorption of this energy twice in a row. The first spike is moreover at the same value as the low population plateau. When observing a grain at this temperature, it is highly unlikely that it is cooling down from a higher temperature, because it would require an additional CR hit. The most probable scenario is that it has just been hit by a CR that deposited the maximal possible energy. Because it has brought its low-temperature ice mantle to a high one, it will very quickly desorb a large amount of molecules, but that will cool it down. Hence, observing a grain at these spikes means they have not desorbed yet because they have not cooled down, explaining why the population is necessarly high. This also explains the negative spikes in the PDF: because the grain cools down so fast thanks to the massive evaporation, it is rarely observed in that state. On the contrary, between the spikes, the grains are more likely to be cooling down from a CR impact that had brought them to a higher energy, and thus to have already lost part of their population.

The population on the grain at rest is just enough for the evaporative cooling to play a role in the first instants following a temperature spike, but is not dominant over all the duration of the cooling of the grain back to equilibrium temperature: it becomes slower than the usual radiative cooling after some of the population has desorbed. The last panel in Figure 17 further attests that where the temperature PDF drops corresponds to where the evaporative timescale is faster than the radiative timescale. The grain cools down faster from these temperatures, so spends less time in these states, resulting in them being less probable.

The iterative method used to solve for the temperature PDF and conditional mean population is slow to converge on this complex transition point. To keep a reasonable computation time, we could not achieve the same level of precision as on other regimes. We thus remark that some small numerical inaccuracies can be noticed.

C. Single-CR approximation of the PDF tail (Derivation of equation (18))

Here we detail the mathematical derivation of the approximation of the PDF tail used in the Single-CR approximation (see 3.3.4). We suppose that the grain was heated to a high energy by a cosmic ray, from which it is cooling down through all the intermediate energies back to equilibrium. We neglect adsorption and photodesorption surface processes. We will treat cooling as a continuous drift (see Appendix E for more mathematical details on what this means). The total cooling rate is $P_{em}^{tot}(E) = P_{em}^{rad}(E) + \Delta E \langle N|E\rangle k_{th}(E)$. The only discrete transition rate left is the absorption rate $R_{abs}(U)$. The master equation with the discrete absorption process and the continuous cooling thus writes:

$$f(E)\int_E^\infty R_{abs}(E'-E)dE' = \int_0^E R_{abs}(E-E')f(E')dE' + \frac{d}{dE}\left(f(E)P_{em}^{tot}(E)\right).$$

As its namesake, the single-CR approximation neglects the possibility of getting hit by another CR during the cooling (and in fact, we neglect the absorption of a photon as well). The left hand side hence becomes zero. Additionally, we know that the bulk of the PDF is peaked around the thermal equilibrium, so we can neglect the low-probability tail and write the integral in the right hand side by approximating the PDF by a Dirac function:

$$\int_0^E R_{abs}(E-E')f(E')dE' \simeq R_{abs}(E-E_{eq}) \simeq R_{abs}(E).$$

This two simplifications yield:

$$R_{abs}(E) = -\frac{d}{dE}\left(f(E)P_{em}^{tot}(E)\right).$$

Integrating from E to ∞ yields (using $f(\infty)=0$):

$$\int_E^\infty R_{abs}(E')dE' = f(E)P_{em}^{tot}(E).$$

Dividing by $P_{em}^{tot}(E)$ we find the wanted approximation:

$$f(E) = \frac{\int_E^\infty R_{abs}(E')dE'}{P_{em}^{rad}(E) + \Delta E \langle N|E\rangle k_{th}(E)}.$$

D. On the second moment approximation

Here we assess the validity of the hypothesis made in Section 3.2.3 which stated that $\langle N^2|E\rangle \simeq \langle N|E\rangle^2$. The hypothesis amounts to saying that the variance of N conditioned on E is zero, i.e. that the process N knowing the thermal energy is deterministic (takes only one value). This hypothesis is only used in the high gas density regime, where evaporative cooling plays a role. In the low gas density regime, the non-linear terms in which we use that assumption are insignificant (since they are evaporative cooling terms). We can thus suppose that we are at high gas density and that the grain cools down mainly by evaporation rather than radiation.

Our goal here is to obtain an analytical approximation of the full two-dimensional PDF for high energies E (since this is where evaporative cooling plays a role), from which we will be able to constrain the conditional variance. We rely on the same method used to estimate the tail of the PDF in the Single-CR approximation 3.3.4, which consists to neglect the less likely processes to obtain an analytically solvable Master Equation. Compared to the single-CR approximation, we additionally suppose radiative cooling to be negligible, and we treat the population as a continuous drift as well.

To summarize the processes, we thus place ourselves at a high energy E and a population N. The population N should necessarily be smaller than the population N_{rest} on the grain when it is at rest at low temperatures (i.e. between CR heating events). The population is only subject to desorption, as a continous drift $-N\,k_{th}(E)$, cooling down the grain as a drift $-P_{em}^{evap}(E,N) = -\Delta E N k_{th}(E)$ These drift expressions are recovered from the Kramers-Moyal expansion (see Appendix E). The grain is brought to high temperatures from equilibrium $E_{eq} \ll E$ at a rate $R_{abs}(E)$. It reaches these energies with its rest population N_{rest}. The master equation for these processes writes:

$$R_{abs}(E)\delta_{N=N_{rest}} = -\frac{d}{dE}\left(\Delta E f(E,N) k_{th}(E) N\right) - \frac{d}{dN}\left(f(E,N) k_{th}(E) N\right)$$

which can be recast as the continuity equation:

$$R_{abs}(E)\delta_{N=N_{rest}} = -\nabla \cdot \left(\begin{bmatrix} \Delta E \\ 1 \end{bmatrix} f(E,N) k_{th}(E) N \right).$$

Since its characteristics are straight lines parallel to the vector $[\Delta E, 1]$, it can be analytically solved:

$$f(E,N) = \frac{R_{abs}(E + \Delta E(N_{rest} - N))}{k_{th}(E) N}.$$

In this expression, $E + \Delta E(N_{rest} - N)$ is the energy the grain had when its population was N_{rest}, i.e. just after the heating event.

We now determine the range $[N_{min},N_{max}]$ of possible values for N at energy E. The highest possible is $N_{max} = N_{rest}$. Since CRs do not deposit more than $\sim 5 \cdot 10^{-13}$J in the grain, $R_{abs} = 0$ above this threshold. This entails that, at energy E, the lowest population value possible is the one on grains that reached the maximum possible energy 510^{-13}J (and had to desorb the most out of their initial N_{rest} population to reach E). This value N_{min} is thus the one such that $E + \Delta E(N_{rest} - N_{min}) \sim 5 \cdot 10^{-13} J$, which yields that the size of the interval of possible populations at energy E is bounded by:

$$N_{rest} - N_{min} = \frac{5 \cdot 10^{-13} - E}{\Delta E} \lesssim 10^6$$

The typical population on the grain verifies $N_{rest} \gg 10^6$ for gas densities n_{gas}(CO) greater than 3cm^{-3}, which covers most of the regime where evaporative cooling plays a role (see 14). This means that we can reasonably neglect the variance of the population N at the energy E in this range of gas densities.

E. Drift-diffusion limit of transition rates (Kramers-Moyal expansion)

Here we present how to approximate a stochastic discrete jump process (such as the stochastic processes E and N we studied in the main text) by a continuous process called a drift-diffusion process. To illustrate the interest for such consideration; one can think of the discrete jumps in grain temperature which happen because of the discrete nature of photons. However, as bigger grains are considered, the photons quanta become negligible, and one is tempted to switch to a more continuous variable and consider those changes as infinitesimal jumps. This can be done mathematically for a general master equation or even just a single term in a transition rate (for instance if one wants to keep photon absorption as discrete jumps but treat cooling as a continuous process). It is the Kramers-Moyal expansion.

Consider an arbitrary transition rate $p_{X\to Y}$ from state X to Y. Let us begin by the simple case of a one-dimension state space $X,Y \in \mathbb{R}$. The full Master Equation for the PDF $f(X)$ might be involving many transition terms modelling different possible physical behaviours, but the only two terms involving $p_{X\to Y}$ are the loss and gain terms. They appear with opposite signs and can be rewritten as integrals over jump lengths rather than states:

$$\int f(Y)p_{Y\to X}dY - f(X)\int p_{X\to Y}dY = \int f(X-Z)p_{X-Z\to X}dZ - f(X)\int p_{X\to X+Z}dZ.$$

These are the terms we wish to build an approximation for, under the hypothesis of small transition jumps. Mathematically, the hypothesis is that in the domain on which the PDF f is not negligible, the jumps $Z = X - Y$ such that $p_{X\to Y}$ is not (nearly) zero must verify $|Z| \ll X,Y$. We can then use a Taylor expansion at X of the function $x \mapsto f(x)p_{x\to x+Z}$, which writes for x close to X:

$$f(x)p_{x\to x+Z} = \sum_{k=0}^{\infty} \frac{(x-X)^k}{k!}\frac{d^k}{dX^k}\left(f(X)p_{X\to X+Z}\right).$$

Plugging this in the above transtion term at $x = X - Z$ yields:

$$\begin{aligned}\int f(X-Z)p_{X-Z\to X}dZ - f(X)\int p_{X\to X+Z}dZ &= \int \sum_{k=0}^{\infty}\frac{(-Z)^k}{k!}\frac{d^k}{dX^k}\left(f(X)p_{X\to X+Z}\right)dZ - f(X)\int p_{X\to X+Z}dZ \\ &= \sum_{k=0}^{\infty}\frac{d^k}{dX^k}\left[f(X)\int\frac{(-Z)^k}{k!}p_{X\to X+Z}dZ\right] - f(X)\int p_{X\to X+Z}dZ\end{aligned}$$

after permutation of the sum of derivatives and the integral sign. Remark now that the $k = 0$ term cancels out with the loss term, so the expansion writes:

$$\int f(X-Z)p_{X-Z\to X}dZ - f(X)\int p_{X\to X+Z}dZ = \sum_{k=1}^{\infty}\frac{d^k}{dX^k}\left[f(X)\int\frac{(-Z)^k}{k!}p_{X\to X+Z}dZ\right].$$

Usually one cuts the expansion at the first or second term, to obtain either a drift (obeying a continuity equation) or a drift-diffusion process (obeying a Fokker-Planck equation). The final result is:

$$\int f(Y)p_{Y\to X}dY - f(X)\int p_{X\to Y}dY = -\frac{d}{dX}\left[f(X)\int Zp_{X\to X+Z}dZ\right] + \frac{1}{2}\frac{d^2}{dX^2}\left[f(X)\int Z^2p_{X\to X+Z}dZ\right].$$

The drift and diffusion terms can be computed once and for all knowing the transition rates.

In higher dimensions, the only difference is the Taylor expansion becoming more complex to write. For the sake of simplicity, we only write it up to $k = 2$ for a smooth function g defined on a subset of $\mathbb{R}^n$:

$$g(\mathbf{X}+\mathbf{Z}) = g(\mathbf{X}) + \mathbf{Z}\cdot\nabla g(\mathbf{X}) + \frac{1}{2}\mathbf{Z}^T\cdot\nabla^{\otimes 2}g(\mathbf{X})\cdot\mathbf{Z} + O(|Z|^3)$$

where $\nabla^{\otimes 2} g$ is the Hessian matrix $\partial_{X_i}\partial_{X_j} g$. Putting the vector $\mathbf{Z}$ under the derivatives transforms the gradients into divergences:

$$g(\mathbf{X}+\mathbf{Z}) = g(\mathbf{X}) + \nabla\cdot(g(\mathbf{X})\mathbf{Z}) + \frac{1}{2}\nabla^{\otimes 2}\cdot(g(\mathbf{X})\mathbf{Z}\otimes\mathbf{Z}) + O(|Z|^3).$$

with last term writing in coordinates:

$$\nabla^{\otimes 2}\cdot(g(\mathbf{X})\mathbf{Z}\otimes\mathbf{Z}) = \sum_{i,j}\partial_{X_i}\partial_{X_j}(gZ_iZ_j).$$

Following the same computations as above, we end up with the approximation:

$$\int f(\mathbf{Y})p_{\mathbf{Y}\to\mathbf{X}}d\mathbf{Y} - f(\mathbf{X})\int p_{\mathbf{X}\to\mathbf{Y}}d\mathbf{Y} = -\nabla\cdot\left[f(\mathbf{X})\int \mathbf{Z}p_{\mathbf{X}\to\mathbf{X}+\mathbf{Z}}d\mathbf{Z}\right] + \frac{1}{2}\nabla^{\otimes 2}\cdot\left[f(\mathbf{X})\int \mathbf{Z}^{\otimes 2}p_{\mathbf{X}\to\mathbf{X}+\mathbf{Z}}d\mathbf{Z}\right].$$

F. Derivations

We detail in these section the derivations that were left out of the main text.

F.1. Derivation of equations (10) and (11)

Our goal is to derive the single-variable equation for the conditional mean population $\langle N|E\rangle$ used in our model, from the 2D Master Equation for (E,N) (8) (in the setting of Section 2, studying photon-induced fluctuations). To do so, we multiply it by N and sum over N, which writes:

$$\sum_{N=0}^{\infty}\int_{\mathbb{R}^+} Nf(E',N)p_{E'\to E}dE' + \sum_{N,N'} Nf(E,N')p_{N'\to N} = \sum_{N=0}^{\infty} Nf(E,N)\int_{\mathbb{R}^+} p_{E\to E'}dE' + \sum_{N,N'} Nf(E,N)p_{N\to N'}$$

The two energy terms can be easily dealt with since the rates $p_{E'\to E}$ do not depend on N, making the quantity $\langle N|E\rangle$ directly appear:

$$\int_{\mathbb{R}^+}\langle N|E'\rangle f(E')p_{E'\to E}dE' + \sum_{N,N'} Nf(E,N')p_{n'\to n} = \langle N|E\rangle f(E)\int_{\mathbb{R}^+} p_{E\to E'}dE' + \sum_{N,N'} Nf(E,N)p_{N\to N'}$$

For the population terms, we will make use of the symmetry of the double sum. Exchanging the dummy variable N' and N in the left sum and passing it over to the right hand side, we get:

$$\int_{\mathbb{R}^+}\langle N|E'\rangle f(E')p_{E'\to E}dE' = \langle N|E\rangle f(E)\int_{\mathbb{R}^+} p_{E\to E'}dE' + \sum_{N,N'}(N-N')f(E,N)p_{N\to N'}$$

Further simplification requires to make use of the explicit expression of the transition rate. Since N' can only be equal to $N+1$ or $N-1$, one gets:

$$\begin{aligned}
\sum_{N,N'}(N-N')f(E,N)p_{N\to N'} &= \sum_{N=0}^{\infty} -f(E,N)p_{N\to N+1} + f(E,N)p_{N\to N-1} \\
&= \sum_{N=0}^{\infty} f(E,N)\left(-k_{ads} + \min(N,N_{act})k_{des}\right) \\
&= \sum_{N=0}^{\infty} f(E,N)\left(-k_{ads} + Nk_{des}\right) \\
&= -f(E)k_{ads} + \langle N|E\rangle f(E)k_{des}
\end{aligned}$$

As stated above, supposing that the PDF tail is negligible beyond N_{act} is what allows us to get rid of the minimum. Plugging this expression in the previous equation gives:

$$\int_{\mathbb{R}^+}\langle N|E'\rangle f(E')p_{E'\to E}dE' = \langle N|E\rangle f(E)\left(\int_{\mathbb{R}^+} p_{E\to E'}dE' + k_{des}\right) - f(E)k_{ads} \tag{21}$$

Integrating over E symmetrizes the two integrals so that they cancel out, and we obtain Equation (11):

$$0 = \int_{\mathbb{R}^+} \langle N|E\rangle f(E) k_{des} dE - k_{ads}$$

We can replace k_{ads} in (21) by this expression to make the equation homogeneous and obtain (10).

$$\langle N|E\rangle = \int_{\mathbb{R}^+} \langle N|E'\rangle \kappa(E,E') dE'$$

where the kernel κ is given by:

$$\kappa(E,E') = \frac{f(E')(p_{E'\to E})}{f(E)M(E)} + \frac{f(E')k_{des}(E')}{M(E)}$$

$$M(E) = k_{des}(E) + \int_{\mathbb{R}^+} p_{E\to F} dF.$$

F.2. Derivation of equation (15)

Our goal is to derive an equation for the marginal temperature PDF $f(E)$ from the master equation for (E,N), this time in the setting of CR-induced fluctuations, so we account for evaporative cooling. The master equation writes, with $X = (E,N)$ and $Y = (E',N')$:

$$\int_{\mathbb{R}^+\times\mathbb{N}} f(Y) p_{Y\to X} dY = f(X) \int_{\mathbb{R}^+\times\mathbb{N}} p_{X\to Y} dY$$

The updated transition rate is:

$$p_{X\to Y} = p^*_{E\to E'}\delta_{N=N'} + p^{a/d}_{N\to N'}\delta_{E=E'} + p^{th}_{X\to Y}$$

The first two are the same as before:

$$p^*_{E\to E'} = \begin{cases} R_{abs}(E'-E) & \text{if } E' > E \\ R_{em}(E-E',E) & \text{if } E' < E \end{cases}$$

and

$$p^{a/d}_{N\to N'} = k_{ads}\delta_{N'=N+1} - N k_{ph}\delta_{N'=N-1}$$

The new rate for thermal desorption affects both energy and population:

$$p^{th}_{X\to Y} = N k_{th}(E)\delta_{N'=N-1}\delta_{E'=E-\Delta E}$$

Developing the master equation with those terms yields:

$$\int_{\mathbb{R}^+} f(E',N) p^*_{E'\to E} dE' + \sum_{N'=0}^{\infty} f(E,N') p^{a/d}_{N'\to N} + f(E+\Delta E,N+1)(N+1)k_{th}(E+\Delta E) = f(E,N)\left[\int_{\mathbb{R}^+} p^*_{E\to E'} dE' + \sum_{N'=0}^{\infty} p^{a/d}_{N\to N'} + N k_{th}(E)\right]$$

Let us first derive the new reduced master equation for the energy marginal PDF by summing over N. Since p^* does not depend on N, the marginal $f(E) = \sum_N f(E,N)$ appears in the radiative terms:

$$\int_{\mathbb{R}^+} f(E') p^*_{E'\to E} dE' + \sum_{N=0}^{\infty}\sum_{N'=0}^{\infty} f(E,N') p^{a/d}_{N'\to N} + \sum_{N=0}^{\infty} f(E+\Delta E,N+1)(N+1)k_{th}(E+\Delta E) = f(E)\int_{\mathbb{R}^+} p^*_{E\to E'} dE' + \sum_{N=0}^{\infty} f(E,N)\sum_{N'=0}^{\infty} p^{a/d}_{N\to N'} + \sum_{N=0}^{\infty} f(E,N) N k_{th}(E)$$

The adsorption/photodesorption summations cancel out, and by factoring k_{th} (which does not depend on population) in front of the other two sums, we are left with:

$$\int_{\mathbb{R}^+} f(E')p^*_{E'\to E}dE' + k_{th}(E+\Delta E)\sum_{N=0}^{\infty} f(E+\Delta E,N+1)(N+1) = f(E)\int_{\mathbb{R}^+} p^*_{E\to E'}dE' + k_{th}(E)\sum_{N=0}^{\infty} f(E,N)N$$

Recognizing the mean population $\langle N|E\rangle$, we have (15):

$$\int_{\mathbb{R}^+} f(E')p^*_{E'\to E}dE' + k_{th}(E+\Delta E)\langle N|E+\Delta E\rangle f(E+\Delta E) = f(E)\int_{\mathbb{R}^+} p^*_{E\to E'}dE' + k_{th}(E)\langle N|E\rangle f(E)$$

This incites us to derive an equation for $\langle N|E\rangle$ as well.

F.3. Derivation of equation (16)

Following what was done above, our goal is to derive an equation for the conditional population $\langle N|E\rangle$ from the master equation for (E,N), when accounting for evaporative cooling. We do so by applying $\sum_N N$ to the master equation. The radiative terms are once again quickly dealt with:

$$\int_{\mathbb{R}^+}\langle N|E'\rangle f(E')p^*_{E'\to E}dE' + \sum_{N=0}^{\infty} N\sum_{N'=0}^{\infty} f(E,N')p^{a/d}_{N'\to N} + \sum_{N=0}^{\infty} Nf(E+\Delta E,N+1)(N+1)k_{th}(E+\Delta E) = \langle N|E\rangle f(E)\int_{\mathbb{R}^+} p^*_{E\to E'}dE' + \sum_{N=0}^{\infty} Nf(E,N)\sum_{N'=0}^{\infty} p^{a/d}_{N\to N'} + \sum_{N=0}^{\infty} Nf(E,N)Nk_{th}(E).$$

Let us regroup the two adsorption sums, the computation is the same as before: we exchange the dummy indices in the second sum to match the $f(E,N')$ of the first one and combine the two:

$$\begin{aligned}\sum_{N=0}^{\infty}\sum_{N'=0}^{\infty} Nf(E,N')p^{a/d}_{N'\to N} - \sum_{N=0}^{\infty}\sum_{N'=0}^{\infty} Nf(E,N)p^{a/d}_{N\to N'} &= \sum_{N=0}^{\infty}\sum_{N'=0}^{\infty} Nf(E,N')p^{a/d}_{N'\to N} - \sum_{N=0}^{\infty}\sum_{N'=0}^{\infty} N'f(E,N')p^{a/d}_{N'\to N}\\ &= \sum_{N=0}^{\infty}\sum_{N'=0}^{\infty}(N-N')f(E,N')p^{a/d}_{N'\to N}\\ &= \sum_{N=0}^{\infty}\sum_{N'=0}^{\infty}(N-N')f(E,N')(k_{ads}\delta_{N=N'+1} - N'k_{ph}\delta_{N=N'-1})\\ &= \sum_{N'=0}^{\infty} f(E,N')k_{ads} - N'f(E,N')k_{ph}\\ &= f(E)k_{ads} - \langle N|E\rangle f(E)k_{ph}.\end{aligned}$$

However, the two other sums do not combine as easily: we can replace N by $N+1$ in the second one since the $N=0$ term is zero, but the PDFs do not have the same energy argument.

$$\begin{aligned}&\sum_{N=0}^{\infty} Nf(E+\Delta E,N+1)(N+1)k_{th}(E+\Delta E) - \sum_{N=0}^{\infty} Nf(E,N)Nk_{th}(E)\\ =&\sum_{N=0}^{\infty} Nf(E+\Delta E,N+1)(N+1)k_{th}(E+\Delta E) - \sum_{N=0}^{\infty}(N+1)f(E,N+1)(N+1)k_{th}(E)\\ =&\sum_{N=0}^{\infty}(N+1)\big(Nf(E+\Delta E,N+1)k_{th}(E+\Delta E) - (N+1)f(E,N+1)k_{th}(E)\big)\end{aligned}$$

We can artificially introduce a $Nf(E,N+1)k_th(E)$ term to cut this sum into the contribution of the $\Delta E = 0$ case and a new one:

$$\sum_{N=0}^{\infty}(N+1)\big(Nf(E+\Delta E,N+1)k_{th}(E+\Delta E)-(N+1)f(E,N+1)k_{th}(E)\big)$$
$$=\sum_{N=0}^{\infty}(N+1)\Big[N\big(f(E+\Delta E,N+1)k_{th}(E+\Delta E)-f(E,N+1)k_{th}(E)\big)+(N-(N+1))f(E,N+1)k_{th}(E)\Big]$$
$$=\Big[\langle N(N-1)|E+\Delta E\rangle f(E+\Delta E)k_{th}(E+\Delta E)-\langle N(N-1)|E\rangle f(E)k_{th}(E)\Big]-\langle N|E\rangle f(E)k_{th}(E)$$

The equation for the population is the same as before with the exception of the new term in square brackets, which involves a finite difference of the second moment of N. This is equation (16):

$$\langle N|E\rangle f(E)\Big(\int_{\mathbb{R}^+}p^*_{E\to E'}dE'+k_{des}(E)\Big)=$$
$$\int_{\mathbb{R}^+}\langle N|E'\rangle f(E')(p^*_{E'\to E}+f(E)k_{des}(E'))dE'+\Big[\langle N(N-1)|\cdot\rangle f k_{th}\Big]_E^{E+\Delta E}$$

G. Details of the Numerical Methods

G.1. Discretization of an integral operator

This section is dedicated to the numerical computation of integral operators of the type

$$\mathcal{L}[f](x)=\int f(y)\kappa(x,y)dy \tag{22}$$

which appear in the equations for both the temperature PDE $f(E)$ and the mean population $\langle N|E\rangle$ (these two functions being eigenvectors of such operators).

Domain and grid. In our applications x and y are energies taking values in $\mathbb{R}^+$. This unbounded domain must be truncated at some energy E_{max} beyond which we suppose the PDF to be zero or negligible. Such a bound can be expected since the absorption (heating) rate is independant of the energy of the grain and becomes small at high photon energies (since the intensity field $I_U(U)$ is zero above the Lyman threshold, so increasing jumps are mostly bounded by this threshold) ; while the emission rate increases with the energy of the grain (as a black body). It is important to fix bounds that will match as best as possible with the decay of the PDF, large enough to capture the full PDF but not too large to not waste the precision of the grid. We can use the equilibrium temperature to estimate the position of the PDF.

A physical lower bound for energies is simply zero. Note that the master equation for $f(E)$ (9) gives a non-zero value for $f(0)$. However, the probability of having zero energies relies on transitions $p_{E'\to 0}$ for which the grain emits all his energy through one photon, this becomes all the more unlikely the bigger the grain. We will thus take E_{min} to be 0 for small grains but a positive value for bigger grains. The discretization grid $\{E_i\}_{1\le i\le I}$ is taken log-uniform, meaning that the $\log E_i$ are evenly spaced (potentially excluding $E_1=0$). We thus let:

$$\text{if } E_1=0,\ \ E_i=E_2\left(\frac{E_I}{E_2}\right)^{\frac{i-2}{I-2}},\ 2\le i\le I$$
$$\text{if } E_1>0,\ \ E_i=E_1\left(\frac{E_I}{E_1}\right)^{\frac{i-1}{I-1}},\ 1\le i\le I$$

Matrix representation of $\mathcal{L}$. We can now consider piece-wise linear functions on the line segments

$[E_i, E_{i+1}]$. The space of such functions being of finite dimension I, the action of $\mathcal{L}$ on this space can be represented as a matrix. Let f be a piece-wise linear function, denoting $f_i = f(E_i)$, its general expression at a point $x \in [E_i, E_{i+1}]$ is obtained by interpolation:

$$f(x) = f_{i+1}\frac{x - E_i}{E_{i+1} - E_i} + f_i\frac{E_{i+1} - x}{E_{i+1} - E_i}$$

Plugging this formula allows us to compute $\mathcal{L}[f]$:

$$\begin{aligned}\mathcal{L}[f](x) &= \sum_{i=1}^{I-1}\int_{E_i}^{E_{i+1}} \kappa(x,y)\left(f_{i+1}\frac{y - E_i}{E_{i+1} - E_i} + f_i\frac{E_{i+1} - y}{E_{i+1} - E_i}\right)dy \\ &= \sum_{i=1}^{I} f_i\left[\int_{E_i}^{E_{i+1}} \kappa(x,y)\frac{E_{i+1} - y}{E_{i+1} - E_i}dy + \int_{E_{i-1}}^{E_i} \kappa(x,y)\frac{y - E_{i-1}}{E_i - E_{i-1}}dy\right]\end{aligned}$$

after reorganizing the sum. Here we set $E_0 = E = 1$ and $E_{I+1} = E_I$ so that the same formula holds even for the first and last terms (cancelling the integrals when needed).
The result $\mathcal{L}[f]$ is surely not a piecewise linear function so we need to project the result onto the space of such functions. The simplest choice is to simply evaluate $\mathcal{L}[f]$ at the E_i with the formula above and interpolate linearly from there. Using the column vector (f_i) to represent piecewise linear functions, the matrix representation of $\mathcal{L}$ is:

$$(\mathcal{L}[f])_i = \sum_{j=1}^{I} L_{ij} f_j$$

$$L_{ij} = \int_{E_j}^{E_{j+1}} \kappa(E_i,y)\frac{E_{j+1} - y}{E_{j+1} - E_j}dy + \int_{E_{j-1}}^{E_j} \kappa(E_i,y)\frac{y - E_{j-1}}{E_j - E_{j-1}}dy$$

We are now left with a finite-dimensional linear system as is common in numerical methods and we can rely on the standard methods of linear algebra.

G.2. Resolution of a finite-dimensional eigenproblem

We wish to solve the eigenproblem $Lf = f$ for eigenvalue 1. Several ways to compute an eigenvector are available. The most straightforward one is to diagonalize the matrix L and find the eigenvector associated to eigenvalue 1. However, we can exploit that the eigenspace for 1 is one-dimensional and that every other eigenvalue is in absolute value less than 1 (a mathematical properties of our stochastic systems that we do not prove here). Iterating the application of the matrix L on f will converge to (a multiple of) the normalized eigenvector, provided our initial guess was not pathological and did not converge to zero. Rather than simply applying L to f at each iteration, the algorithm we used is the following: we start from a constant initial guess f_0, and we set $L_0 = L$. We then iterate the following loop until a given precision is achieved for the error $\|f_{k+1} - f_k\|$ (in practice 10^{-15}):

- Compute $L_{k+1} = L_k^2$
- Set $f_{k+1} = L_{k+1} f_k$
- Normalize both f_{k+1} and L_{k+1} by $\|f_{k+1}\|$

This is more efficient than the naive algorithm even though we perform matrix-matrix multiplications. Indeed, to converge, we need to do a large number n of iterations, $n \gg I$. Iterating L on f amounts to doing n matrix-vector multiplications, for a complexity $O(nI^2)$. Our method has complexity $O(I^3)$ per iteration because of the matrix-matrix multiplication. However we iterate only $\log_2 n$ times, for a total complexity $O(\log_2 nI^3)$. Our method was faster since in this setting we had $\log_2 nI < n$.

G.3. Iterative method for the non-linear system {(15), (16)}

Here we present the iterative method used to solve the non-linear system in section 3.2.3. We begin by recasting the equations (15) and (16) as eigenproblems, to be solved with the method described above.

Equation for $f(E)$. We recall (15):

$$\int_{\mathbb{R}^+} f(E')p^{rad}_{E'\to E}dE' + k_{th}(E+\Delta E)\langle N|E+\Delta E\rangle f(E+\Delta E) = f(E)\int_{\mathbb{R}^+} p^{rad}_{E\to E'}dE' + k_{th}(E)\langle N|E\rangle f(E)$$

It can be cast into an eigenproblem

$$\mathcal{L}[f](E) := \int f(E')\kappa(E,E')dE' = f(E)$$

if one defines the kernel κ^E (assuming knowledge of $\langle N|E\rangle$)

$$\kappa^E(E,E') := \frac{p^{rad}_{E'\to E} + k_{th}(E')\langle N|E'\rangle\delta(E'-E-\Delta E)}{\lambda(E)}$$

where the loss λ is the integral over E of the numerator:

$$\lambda(E) = \int_{\mathbb{R}^+} p^{rad}_{E\to F}dF + k_{th}(E)\langle N|E\rangle.$$

This means that, assuming knowledge of $\langle N|E\rangle$, we can rely on the linear algebra methods described above to solve for f.

Equation for $g(E)$. We let $g(E) = \langle N|E\rangle f(E)$, which reduces the non-linearity of the equations. We recall (16) in terms of g:

$$\int_{\mathbb{R}^+} g(E')(p^*_{E'\to E} + f(E)k_{des}(E'))dE' + \langle N-1|E+\Delta E\rangle_+ g(E+\Delta E)k_{th}(E+\Delta E) = g(E)\left(\int_{\mathbb{R}^+} p^*_{E\to E'}dE' + k_{des}(E)\right) + \langle N-1|E\rangle_+ g(E)k_{th}(E)$$

where we have used the approximation of the second moment (Eq. (17)), and where $\langle N-1|E\rangle_+ = \max(\langle N-1|E\rangle,0)$. The eigenproblem writes

$$\tilde{\mathcal{L}}[g](E) := \int g(E')\kappa^N(E,E')dE' = g(E)$$

with the kernel

$$\kappa^N(E,E') := \frac{p^{rad}_{E'\to E} + f(E)k_{des}(E') + k_{th}(E')\langle N-1|E'\rangle_+\delta(E'-E-\Delta E)}{\mu(E)}$$

and the loss term

$$\mu(E) = \int_{\mathbb{R}^+} p^{rad}_{E\to F}dF + k_{des}(E) + k_{th}(E)\langle N-1|E\rangle_+.$$

Assuming knowledge of f and $\langle N|E\rangle$, we can solve for g.

Iteration loop. The coupled system was solved by iterating on $(f,g,\langle N|E\rangle)$ as follows. We began with no population $\langle N|E\rangle = 0$. Using the algorithm from G.2, we can solve for the eigenproblem for f (under the hypothesis $\langle N|E\rangle = 0$, the solution we obtain is simply the PDF without evaporative cooling). The knowledge of f allows to solve the eigenproblem for g. Knowing f and g give a value $\langle N|E\rangle = \frac{g}{f}$. We can plug it in the kernel κ^E to solve for a new f, a new g, a new population, and so on. We stop iterating when the norm of the difference between the functions at iteration k and iteration $k+1$ are smaller than a certain threshold (10^{-7}).